Daily Mean Reversion in Global Equity Indices: Serial Dependence, Tail Risk, and Exploitation

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Abstract

This paper examines daily mean reversion across six major equity indices in North America, Europe, and Asia-Pacific spanning the period 2002–2026 (over 34,000 trading sessions). Linear serial dependence tests reject weak-form market efficiency, revealing significant negative first-order autocorrelation alongside heavy-tailed, negatively skewed residuals. To exploit this anomaly while controlling downside exposure and secular market drift, we introduce two systematic trading architectures: an unfiltered symmetric baseline and an asymmetric, regime-filtered model. Rather than relying on nominal compounding, strategy evaluation highlights pronounced statistical robustness and geographical invariance across all three continents. An asymmetric 200-day moving average filter suppresses counter-trend short entries, elevating the cross-sectional Sharpe ratio to 1.742 and compressing maximum drawdown to −18.95% (compared to −56.68% for the passive benchmark), alongside an improved Profit Factor of 1.498. Furthermore, a strict 0.50% intraday stop-loss effectively truncates left-tail risk, inducing strong positive return skewness (+2.513) and confirming the structural stability and institutional feasibility of daily mean reversion across global financial centers.

Keywords: Mean Reversion, Systematic Trading, Market Efficiency, Regime Filter, Tail Risk, Serial Correlation.
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1 Introduction

The cornerstone of modern financial economics is the Efficient Market Hypothesis (EMH), which posits that asset prices fully reflect all available information and that systematic risk-adjusted excess returns cannot be extracted from public price signals (Fama, 1970, 1991). In its weak-form interpretation, the EMH implies that asset returns follow a random walk or martingale process, exhibiting negligible linear serial dependence at short horizons. However, an extensive empirical literature has challenged this benchmark. Lo and MacKinlay (1988) applied variance ratio tests to U.S. stock returns and documented significant positive serial correlation at weekly intervals, rejecting the uncorrelated-increments assumption inherent in the random-walk hypothesis. Conversely,

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examining daily and short-horizon equity dynamics, Jegadeesh (1990) identified statistically significant negative first-order serial correlation and documented substantial short-term mean reversion. Subsequently, Brock et al. (1992) demonstrated that technical trading rules, such as moving averages and trading-range breakouts, generated statistically significant predictive power. Crucially, their findings revealed a fundamental directional asymmetry: buy signals yielded positive average returns with lower volatility, whereas sell signals produced negative returns accompanied by higher volatility. In an international equity context, this line of inquiry has been corroborated by Metghalchi et al. (2008), who show that moving-average trading rules can exploit persistent market inefficiencies and generate superior risk-adjusted returns relative to passive benchmarks. Furthermore, in a comprehensive international survey, Amini et al. (2013) synthesize the evidence on short-term price predictability conditional on prior price shocks, highlighting that market microstructure frictions, order-flow imbalances, and behavioral overreaction systematically drive short-horizon mean-reverting dynamics across global markets.

Beyond serial dependence, asset returns systematically violate Gaussian distribution assumptions (Markowitz, 1952; Black and Scholes, 1973). Empirical return series consistently display fat tails, excess kurtosis, negative skewness, and volatility clustering, where high-volatility events cluster together over time (Mandelbrot, 1963; Fama, 1965; Engle, 1982; Cont, 2001). Moreover, regime-dependent return dynamics indicate that short-horizon predictability varies across macro states (McQueen and Thorley, 1991). In broad equity indices, these stylized facts are compounded by a positive long-run structural drift driven by the equity risk premium. Consequently, symmetric short-term mean-reversion strategies encounter fundamentally asymmetric risk-return profiles: short positions taken against the prevailing bull market face severe tail losses and adverse structural drift, whereas long positions benefit from long-term market growth.

To evaluate whether short-term mean reversion constitutes an invariant structural property of modern equity markets rather than an in-sample artifact of the U.S. large-cap universe, this paper presents a cross-sectional empirical investigation across six major global equity indices spanning North America (S&P 500, Russell 2000), Europe (FTSE 100, EURO STOXX 50), and the Asia-Pacific region (Nikkei 225, S&P/ASX 200) over a 24-year horizon from 2002 to 2026 (encompassing over 34,000 aggregate daily trading sessions). Across regional markets, standard linear martingale hypotheses are evaluated via lag-1 Ljung-Box Q-tests (Ljung and Box, 1978) and $AR(1)$ specifications estimated via Newey and West (1987) heteroskedasticity and autocorrelation-consistent (HAC) standard errors. Across the global panel, underlying benchmark buy-and-hold returns exhibit severe deviations from normality, featuring an equal-weighted cross-sectional average excess kurtosis of +8.42, negative skewness (-0.255), and pronounced left-tail vulnerability, as evidenced by a cross-sectional 95% Value-at-Risk ($VaR_{95\%}$) of -1.95% and 95% Conditional Value-at-Risk ($CVaR_{95\%}$) of -3.04%. Formal residual diagnostics decisively reject Gaussianity via the Jarque-Bera test (Jarque and Bera, 1980), demonstrating that unhedged global mean reversion is universally exposed to asymmetric downside tail risk.

Motivated by these econometric findings, this paper introduces two algorithmic architectures designed to systematically exploit short-term mean reversion while addressing tail risk and structural market drift. As a first primary contribution, we formulate an unfiltered *symmetric*

mean-reversion strategy combining daily single-candle reversal signals with a strict 0.50% intraday stop-loss. This baseline operates agnostic to prevailing market trend: long positions are triggered after down-candles ($Close_t < Open_t$) and short positions after up-candles ($Close_t > Open_t$). By applying uniform execution across both market directions, this benchmark isolates the intrinsic strength of cross-regional daily reversals and demonstrates how programmatic stop-loss truncation reshapes return distribution tails. As a second and refined contribution, we develop an *asymmetric*, regime-conditioned extension engineered to resolve the structural friction imposed by the equity risk premium. Building directly upon the symmetric model, the asymmetric architecture integrates a 200-period Simple Moving Average (SMA_{200}) filter: long entries remain unconditional to continuously harvest positive secular drift, whereas short entries are restricted exclusively to regimes where closing prices trade at or below the trendline ($Close_t \leq SMA_{200,t}$). This conditioning prunes counter-trend short entries during secular expansions, reducing the total trade count by approximately 36.6% (executing 3,669 positions versus 5,790 in the symmetric baseline) to achieve superior risk containment while maintaining high profitability.

An empirical backtest across the six global indices over the 2002–2026 sample period demonstrates substantial risk-return efficiency gains relative to the passive buy-and-hold benchmark, which suffered from severe tail risk and elevated variance (annualized volatility of 20.28%, a Sharpe ratio of 0.340, and an average maximum drawdown of −56.68%). Crucially, the primary efficacy of both implementations resides in variance compression and tail reshaping rather than raw nominal accumulation. The baseline symmetric architecture establishes a superior risk-adjusted profile, yielding an average Sharpe ratio of 1.640, a Sortino ratio of 7.044, and a Calmar ratio of 1.340, while halving benchmark maximum drawdown to −26.84%. The asymmetric refinement delivers further structural variance dampening: by selectively suppressing counter-trend short trades (eliminating an average of 2,122 executions per market), it compresses annualized volatility from 15.27% down to 13.66% and tightens maximum drawdown to −18.95%. This variance reduction translates into an elevated Sharpe ratio of 1.742, a Calmar ratio of 1.653, and an improved trade-level Profit Factor (1.498 vs. 1.347). Underpinning these risk-adjusted gains, the 0.50% intraday stop-loss mechanism structurally truncates negative tail realizations, transforming the empirical distribution from heavy negative skewness into pronounced positive asymmetry (+2.544 in the symmetric baseline and +2.513 in the asymmetric model), thereby producing stable, normalized excess returns across diverse macro regimes.

2 Market Analysis & Price Dynamics

To corroborate the theoretical framework and formalize the empirical boundaries of the Efficient Market Hypothesis (EMH), this section presents an econometric analysis of daily return dynamics across global equity indices. The objective is threefold: (i) quantify the magnitude of short-term serial dependence; (ii) establish a formal predictive bridge between execution-level candle signals, macro regime filters, and subsequent returns; and (iii) diagnose the severe distributional departures from Gaussianity that govern tail risk. The empirical analysis spans a 24-year continuous horizon from 2002 to 2026 (6,035 trading sessions for the benchmark S&P 500 and over 34,000 aggregate cross-sectional observations) across six developed market benchmarks: the S&P 500 and Russell

2000 (North America), the FTSE 100 and EURO STOXX 50 (Europe), and the Nikkei 225 and S&P/ASX 200 (Asia-Pacific). Throughout this analysis, cross-sectional equal-weighted averages establish the systemic properties of daily mean reversion across international financial centers.

2.1 Linear Serial Dependence and Rejection of the Random Walk

The initial stage of empirical validation evaluates the presence of linear serial correlation in daily logarithmic close-to-close returns, defined as

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right), \tag{1}$$

where $P_{i,t}$ denotes the adjusted closing price of index i on day t. Under the weak-form EMH, price increments follow a martingale difference sequence wherein historical information yields zero linear forecastability:

$$\mathrm{Cov}(r_{i,t}, r_{i,t-k}) = 0, \qquad \forall\, k \neq 0. \tag{2}$$

We evaluate the joint absence of serial correlation across short horizons via the portmanteau test of Ljung and Box (1978):

$$Q(m) = n(n+2)\sum_{k=1}^{m}\frac{\hat{\rho}_k^2}{n-k}, \tag{3}$$

where n is the sample size, m is the maximum lag length, and $\hat{\rho}_k$ represents the sample autocorrelation at lag k.

For the first lag ($m = 1$), the S&P 500 decisively rejects the uncorrelated-increments null hypothesis with a test statistic of $Q(1) = 82.40$ ($p < 0.001$). Across the full international cross-section, the equal-weighted average statistic stands at $\bar{Q}(1) = 36.39$ ($p = 0.0434$). These findings indicate that the daily covariance matrix of international equities is statistically distinguishable from a diagonal matrix, driven by microstructure frictions including bid-ask bounce (Roll, 1984), inventory balancing, and transitory order-flow imbalances (Lo and MacKinlay, 1988; Jegadeesh, 1990; Cont, 2001; Amini et al., 2013).

2.2 Linear AR(1) Dynamics and the Necessity of a Predictive Bridge

To quantify the magnitude and direction of this linear dependence, an autoregressive specification of order one, AR(1), is estimated for each index:

$$r_{i,t} = \alpha_i + \phi_i r_{i,t-1} + \varepsilon_{i,t}, \tag{4}$$

where α_i is the unconditional drift, ϕ_i is the first-order slope parameter, and $\varepsilon_{i,t}$ is the error disturbance. Estimation is conducted via Ordinary Least Squares with Newey and West (1987) heteroskedasticity and autocorrelation-consistent (HAC) standard errors (1-lag truncation).

As documented in Table 1, the autoregressive parameter for the S&P 500 is negative and highly statistically significant ($\phi = -0.119$, HAC $t = -3.879$, $p = 0.0001$). However, an inspection of the broader international cross-section reveals critical empirical qualifications:

1. **Statistical Fragility in Peripheral Markets:** While the cross-sectional average autoregressive

Table 1. Cross-Sectional AR(1) Model Diagnostics and Serial Dependence

Metric	S&P 500	Benchmark Mean	Cross-Sectional Range
Autoregressive Coeff. (ϕ)	-0.119	-0.069	$[-0.119, -0.014]$
Ljung-Box $Q(1)$	82.40	36.39	$[1.09, 82.40]$
p-value (Ljung-Box)	< 0.001	0.0434	$[< 0.001, 0.2960]$
R^2	0.0142	0.0063	$[0.0002, 0.0142]$
Adjusted R^2	0.0141	0.0061	$[0.0000, 0.0141]$
HAC t-statistic	-3.879	-2.478	$[-3.879, -0.759]$
p-value (t-test)	0.0001	0.0964	$[0.0001, 0.4480]$
Durbin–Watson Statistic	2.001	1.999	$[1.997, 2.001]$
AIC	$-34{,}984.04$	$-34{,}678.93$	$[-36{,}950.27, -31{,}779.99]$
BIC	$-34{,}970.71$	$-34{,}665.61$	$[-36{,}936.94, -31{,}766.73]$

Notes: Estimation results for the autoregressive model $r_{i,t} = \alpha_i + \phi_i r_{i,t-1} + \varepsilon_{i,t}$ across the six global equity indices over the 2002–2026 sample period. Statistical inference is conducted using Newey and West (1987) HAC standard errors with 1 lag. The 'Benchmark Mean' column reports the cross-sectional equal-weighted arithmetic mean across all six markets (S&P 500, Russell 2000, FTSE 100, EURO STOXX 50, Nikkei 225, S&P/ASX 200).

slope is negative ($\bar{\phi} = -0.069$, $t = -2.478$), unconditional linear predictability weakens substantially outside U.S. large-cap equities. Across the six indices, the HAC t-statistic attenuates to as low as -0.759 ($p = 0.4480$), producing a cross-sectional mean p-value of 0.0964. In multiple jurisdictions, the null hypothesis $\phi_i = 0$ cannot be rejected at conventional significance levels ($\alpha = 0.05$).

2. **Modest Explanatory Variance:** The empirical coefficient of determination averages $\bar{R}^2 = 0.0063$ (0.0142 for the S&P 500), indicating that unconditional close-to-close lags account for less than 1% of one-day-ahead return variation.

3. **Methodological Disconnection with Execution Signals:** Crucially, the standard AR(1) framework estimates serial correlation strictly on close-to-close returns ($r_{i,t-1}$). In contrast, systematic mean-reversion trading architectures do not condition entries on total 24-hour returns; rather, they exploit the directional polarity of intraday cash sessions, entering on $\text{sgn}(Close_{i,t} - Open_{i,t})$. A model relying strictly on close-to-close lags obscures the structural interaction between intraday auction imbalances and macro trend regimes.

2.2.1 Formalizing the Predictive Candle-Regime Interaction Model

To resolve this methodological friction and econometric disconnect, we formulate an explicit predictive regression model mapping the operational execution signal and the secular regime filter onto forward daily returns:

$$r_{i,t+1} = \alpha_i + \beta_{1,i} \cdot \text{CandleDir}_{i,t} + \beta_{2,i} \cdot D_{i,t} + \beta_{3,i} \cdot (\text{CandleDir}_{i,t} \times D_{i,t}) + \eta_{i,t+1}, \tag{5}$$

where:

- $r_{i,t+1} = \ln(Close_{i,t+1}/Close_{i,t})$ is the forward daily return earned by an overnight position held through day $t + 1$;

- CandleDir$_{i,t} \in \{-1, +1\}$ captures intraday session polarity, defined as +1 if $Close_{i,t} > Open_{i,t}$ (up-candle) and −1 if $Close_{i,t} < Open_{i,t}$ (down-candle);
- $D_{i,t} = \mathbf{1}_{\{Close_{i,t} > SMA_{200,i,t}\}} \in \{0, 1\}$ is an indicator variable identifying secular bull regimes where closing price exceeds the 200-day Simple Moving Average (SMA_{200});
- CandleDir$_{i,t} \times D_{i,t}$ is the interaction term isolating the marginal impact of intraday polarity conditional on secular upward momentum.

Model (5) is estimated via OLS with Newey and West (1987) HAC covariance estimators (1-lag truncation) over the 2002–2026 sample period. The empirical results are presented in Table 2.

Table 2. Predictive Bridge Regression Diagnostics: Candle Polarity and Trend Interaction

Index	α	β_1	$t(\beta_1)$	$p(\beta_1)$	β_2	β_3	$t(\beta_3)$	$p(\beta_3)$
S&P 500	0.00019	−0.00156	−3.169	0.0015	0.00014	0.00128	2.514	0.0119
Russell 2000	0.00024	−0.00153	−3.165	0.0016	0.00002	0.00138	2.689	0.0072
FTSE 100	0.00034	−0.00021	−0.603	0.5462	−0.00034	0.00005	0.128	0.8981
EURO STOXX 50	−0.00002	−0.00052	−1.216	0.2239	0.00017	0.00038	0.808	0.4189
Nikkei 225	0.00028	−0.00032	−0.853	0.3939	−0.00004	0.00021	0.508	0.6117
S&P/ASX 200	−0.00003	−0.00076	−2.351	0.0188	0.00026	0.00061	1.787	0.0740
Benchmark Mean	**0.00017**	**−0.00082**	**−1.893**	**0.1977**	**0.00003**	**0.00065**	**1.406**	**0.3370**

Notes: Estimation results for the interaction specification $r_{i,t+1} = \alpha_i + \beta_{1,i}\text{CandleDir}_{i,t} + \beta_{2,i}D_{i,t} + \beta_{3,i}(\text{CandleDir}_{i,t} \times D_{i,t}) + \eta_{i,t+1}$ spanning 2002–2026. Standard errors and t-statistics are computed via Newey and West (1987) HAC covariance matrices (1 lag). CandleDir$_{i,t} = \text{sgn}(Close_{i,t} - Open_{i,t})$; $D_{i,t} = \mathbf{1}_{\{Close_{i,t} > SMA_{200,i,t}\}}$. The cross-sectional average R^2 across the six markets is 0.19% (ranging up to 0.46% on the S&P 500).

The empirical estimates in Table 2 provide two crucial econometric insights that directly validate the algorithmic architecture:

1. **Intraday Reversal Significance in Bear Regimes ($\beta_1 < 0$):** In secular bear regimes ($D_{i,t} = 0$), the marginal relationship between candle polarity and subsequent return is governed strictly by β_1. The parameter is universally negative across all six markets, reaching high statistical significance in North American and Australian markets (e.g., $\beta_1 = -0.00156$, $t = -3.169$ for the S&P 500; $\beta_1 = -0.00153$, $t = -3.165$ for the Russell 2000; and $\beta_1 = -0.00076$, $t = -2.351$ for the S&P/ASX 200). This negative slope confirms that when prices trade below their 200-day moving average, up-candles (CandleDir = +1) systematically predict negative forward returns, whereas down-candles (CandleDir = −1) precede robust positive reversals.

2. **Destructive Drift Interaction for Short Positions ($\beta_3 > 0$):** In secular bull regimes ($D_{i,t} = 1$), the total marginal impact of an intraday candle is $\beta_{1,i} + \beta_{3,i}$. Across all six indices without exception, β_3 is strictly positive ($\bar{\beta}_3 = +0.00065$), with statistically significant estimates on the S&P 500 ($\beta_3 = +0.00128$, $t = 2.514$, $p = 0.0119$) and Russell 2000 ($\beta_3 = +0.00138$, $t = 2.689$, $p = 0.0072$).

From an asset-pricing perspective, a positive interaction parameter $\beta_3 > 0$ exerts a counteracting effect against $\beta_1 < 0$. Following an up-day (CandleDir = +1) during a secular expansion ($D_{i,t} = 1$),

the expected return on the underlying index increases by $\beta_1 + \beta_3$ (offsetting the negative reversal by 82% on the S&P 500 and 90% on the Russell 2000). Consequently, establishing a short position against an up-candle in a bull market subjects the portfolio to severe positive drift and frictional drag. Conversely, following a down-candle (CandleDir = -1), long positions benefit additively from both the mean-reverting bounce and the structural equity risk premium ($\beta_2 > 0$).

This empirical finding supplies the formal econometric justification for our *asymmetric regime filter*: by suppressing short entries whenever $Close_{i,t} > SMA_{200,i,t}$, the strategy eliminates trades where β_3 erodes reversal margins, while preserving unconditional long entries to harvest the secular equity drift.

2.3 Residual Diagnostics and Deviations from Gaussian Normality

The concluding stage of the market analysis evaluates the higher-order standardized moments and residual distributions from the baseline specifications. Canonical quantitative paradigms—from mean-variance portfolio theory (Markowitz, 1952) to classical derivative pricing (Black and Scholes, 1973)—presuppose normally distributed return innovations. In contrast, empirical financial time series systematically display negative skewness, leptokurtosis, and extreme volatility clustering (Mandelbrot, 1963; Fama, 1965; Cont, 2001).

Table 3. Cross-Sectional Residual Distribution Diagnostics

Diagnostic Test / Metric	S&P 500	Benchmark Mean	Gaussian Benchmark
Skewness	−0.609	−0.508	0.000 (Symmetric)
Kurtosis (Pearson)	15.459	12.920	3.000 (Mesokurtic)
Excess Kurtosis	+12.459	+9.920	0.000
Omnibus Normality Stat	1,449.44	1,227.09	$\chi^2(2)$ asymptotic
p-value (Omnibus)	< 0.001	< 0.001	$p > 0.05$ (Fail to reject)
Jarque–Bera (JB) Stat	37,705.23	25,555.75	$\chi^2(2)$ asymptotic
p-value (Jarque–Bera)	< 0.001	< 0.001	$p > 0.05$ (Fail to reject)
Durbin–Watson Statistic	2.001	1.999	2.000 (Zero serial corr.)
Condition Number	84.42	83.43	< 30.00 (Collinearity bound)

Notes: Residual diagnostics obtained from AR(1) regressions with Newey and West (1987) HAC covariance matrices across the 2002–2026 sample period. Kurtosis follows Pearson's convention (Gaussian = 3.000). The 'Benchmark Mean' column denotes the equal-weighted cross-sectional mean across the six global indices (S&P 500, Russell 2000, FTSE 100, EURO STOXX 50, Nikkei 225, S&P/ASX 200).

The diagnostic metrics reported in Table 3 demonstrate severe violations of normality across all financial centers:

- **Pronounced Negative Asymmetry:** Empirical residual skewness settles at −0.609 for the S&P 500 and averages −0.508 across the international panel. This structural left-tail elongation reflects the fact that market liquidations occur with substantially greater velocity and amplitude than upward drift. In unhedged mean-reversion trading, entering positions counter to persistent downward moves exposes capital to severe negative tail risk.
- **Extreme Leptokurtosis:** The fourth standardized moment displays severe fat tails. Residual kurtosis reaches 15.459 (excess kurtosis +12.459) for the S&P 500 and averages 12.920 (excess

kurtosis +9.920) across the cross-sectional benchmark, massively exceeding the Gaussian baseline of 3.0.

- **Decisive Rejection of Normality:** Joint hypothesis tests decisively reject Gaussianity. The Omnibus test yields an average statistic of 1,227.09 ($p < 0.001$), while the Jarque and Bera (1980) test yields an empirical statistic of 37,705.23 on the S&P 500 and a cross-sectional mean of 25,555.75 ($p < 0.001$).
- **Absence of Remaining First-Order Dependence:** The Durbin–Watson statistic averages 1.999 (2.001 for the S&P 500), confirming that the first-order autoregressive terms absorb linear dependence, leaving disturbance terms serially uncorrelated.

Taken together, these empirical regularities establish that raw daily equity returns exhibit statistically detectable but bounded linear predictability, coupled with extreme non-normal downside vulnerability. A naive symmetric mean-reversion rule is consequently structurally incomplete: achieving economic viability requires (i) an asymmetric macro trend filter (SMA_{200}) to neutralize the destructive β_3 interaction during secular bull markets, and (ii) a strict intraday stop-loss mechanism to structurally truncate the negative tail distribution highlighted by residual leptokurtosis.

3 Theoretical Framework & Analytical Model

The econometric findings established in the preceding section provide the empirical motivation for defining an analytical reference framework of short-horizon equity return dynamics. Rather than postulating a novel stochastic theory, the objective is to establish a tractable mathematical benchmark that formally disentangles the structural long-term secular drift from transient, high-frequency mean-reverting pressures.

The preliminary diagnostics demonstrate that daily equity index returns cannot be characterized as pure martingale difference sequences. Specifically, the statistically significant first-order serial dependence detected across North America, Europe, and Asia-Pacific confirms the presence of a persistent, localized mean-reverting force. Nevertheless, the low explanatory power documented across indices ($R^2 = 0.0142$ for the S&P 500, with a cross-sectional pooled average of $R^2 = 0.0063$) indicates that this predictable component accounts for only a minor fraction of daily return variance. Consequently, price trajectories remain predominantly stochastic, and asset pricing efficiency is preserved in the broad sense. The purpose of the analytical framework is therefore to formally isolate the structural trade-off between the secular equity risk premium and transient local corrections within an overwhelmingly noisy environment.

To model this interaction explicitly, the underlying conditional return generation process is parameterized as follows:

$$R_t = \mu - \theta(R_{t-1} - \mu) + \epsilon_t, \tag{6}$$

where R_t denotes the continuously compounded return on trading session t, $\mu > 0$ represents the unconditional long-run equilibrium drift of the equity market, $\theta > 0$ measures the speed of short-term mean reversion, and ϵ_t represents the stochastic innovation term.

Equation (6) does not represent a new asset pricing model; rather, it constitutes a structural reparameterization of a canonical first-order autoregressive AR(1) process around a non-zero mean:

$$R_t = (1 + \theta)\mu - \theta R_{t-1} + \epsilon_t. \tag{7}$$

In this formulation, the conventional autoregressive coefficient ϕ maps algebraically to $-\theta$. An estimated negative autoregressive slope ($\phi < 0$) corresponds directly to a strictly positive feedback intensity ($\theta > 0$). This parameterization provides a clear economic decomposition: whenever the previous return deviates from equilibrium ($R_{t-1} \neq \mu$), the corrective term $-\theta(R_{t-1} - \mu)$ exerts a compensatory pull toward the secular mean. If $R_{t-1} > \mu$, the conditional expectation is damped; conversely, if $R_{t-1} < \mu$, the conditional expectation is augmented.

Empirically, this benchmark mirrors the AR(1) estimations obtained across the global panel, where the pooled autoregressive coefficient averages $\bar{\phi} = -0.069$ ($\bar{\theta} = 0.069$, HAC t-statistic $= -2.478$) and reaches $\phi = -0.119$ ($t = -3.879$, $p = 0.0001$) on the S&P 500. Crucially, because linear predictability explains approximately 1% of return variation, between 98.58% and 99.37% of daily price behavior is driven by the innovation ϵ_t. The reference model in equation (6) is therefore an analytical apparatus designed to isolate a modest conditional edge embedded in a predominantly stochastic data-generating process.

3.1 Non-Gaussian Innovations and Tail Risk

A critical dimension of the analytical framework concerns the properties of the disturbance term ϵ_t. Econometric residual diagnostics decisively demonstrate that the innovations depart from Gaussian assumptions across all major equity indices. For the baseline S&P 500, the estimated residual skewness is -0.609 and Pearson kurtosis is 15.459 (excess kurtosis of +12.459). Across the broader international cross-section, pooled residual skewness averages -0.508 and kurtosis reaches 12.920. Formal Omnibus and Jarque–Bera tests (37,705.23 on the S&P 500; 25,555.75 pooled) reject the null hypothesis of normality at all conventional significance levels.

These empirical properties demonstrate that ϵ_t must be evaluated as a heavy-tailed, negatively skewed process. The presence of pronounced leptokurtosis indicates that extreme realization probabilities decay far more slowly than implied by a Gaussian law (Mandelbrot, 1963; Fama, 1965; Cont, 2001). For trading models predicated on mean reversion, this represents an acute structural risk: an operational strategy may capture frequent, moderate gains as prices revert toward μ, yet remain vulnerable to catastrophic losses caused by realizations located in the deep tails of ϵ_t.

The negative skewness further compounds this operational hazard, indicating that severe negative drawdowns occur with greater magnitude and frequency than large positive expansions. Residual Durbin–Watson statistics (2.001 for the S&P 500 and 1.999 pooled) confirm that the linear autoregressive component exhausts first-order temporal dependence (Durbin and Watson, 1951). However, the residual non-normality underscores that higher-order moments remain active, establishing a theoretical requirement for explicit risk-truncation mechanisms (e.g., algorithmic stop-losses) rather than relying solely on unconditional expected values.

3.2 Equilibrium Return and the Mean-Reverting Trade-Off

The parameter μ defines the unconditional macroeconomic drift that compensates investors for bearing equity risk. Because productive capital allocation commands a positive expected return over extended horizons, the assumption that $\mu > 0$ is a non-trivial structural feature of developed financial markets.

Isolating μ from the mean-reversion parameter θ in equation (6) resolves a common conceptual ambiguity: a negative autoregressive coefficient does not imply convergence toward zero. Instead, short-term shocks oscillate around a positive, upward-sloping equilibrium trajectory. Imposing a zero-mean assumption would confound two fundamentally distinct market phenomena:

1. The secular, macroeconomic upward drift driven by economic growth and capital accumulation ($\mu > 0$);

2. The high-frequency, transient price reversal driven by liquidity consumption and temporary order-flow imbalances ($\theta > 0$).

When θ is small, return dynamics are dominated by secular growth μ and stochastic white noise ϵ_t. As θ increases, the magnitude of the transient correction relative to prior deviations grows. The central analytical challenge in designing systematic mean-reversion systems lies in identifying market states where these two forces align constructively rather than counteract one another.

3.3 Regime Dependence and the Macroeconomic Filter

Although the stationary specification in equation (6) provides an accurate representation of unconditional average dependence, assuming constant parameters across distinct market environments is overly restrictive (McQueen and Thorley, 1991). In particular, the economic efficacy of a mean-reverting signal depends heavily on whether the underlying market regime reinforces or opposes the expected price recovery.

To formalize this regime dependency without introducing spurious parameters, we implement a state indicator based on the 200-period Simple Moving Average (SMA_{200}):

$$D_t = \begin{cases} 1, & \text{if } Close_t > SMA_{200,t}, \\ 0, & \text{if } Close_t \leq SMA_{200,t}. \end{cases} \tag{8}$$

When $D_t = 1$, the equity market resides in a secular expansionary regime; when $D_t = 0$, it resides in a corrective or non-positive regime.

The interaction between regime state D_t and short-term mean reversion can be formalized as an alignment problem between two drift components: the local corrective drift $-\theta(R_{t-1} - \mu)$ and the macro trend state represented by D_t. In a bull regime ($D_t = 1$), an intraday downward shock generates a mean-reverting signal whose anticipated upside bounce is supported by secular drift $\mu > 0$. In contrast, an unconditioned short trade executed during a bull regime creates direct friction between the temporary downward expectation and the underlying upward drift. This theoretical mechanism rationalizes the inclusion of D_t as a structural filter to prune counter-trend short entries during secular bull markets (Metghalchi et al., 2008; Gerritsen, 2016).

3.4 Asymmetry Between Long and Short Reversal Signals

The core theoretical contribution of this section lies in formalizing the structural asymmetry between positive and negative reversal signals. Standard linear specifications implicitly impose directional symmetry: a positive deviation is assumed to trigger a corrective pull identical in relative magnitude and statistical likelihood to that triggered by a negative deviation. However, the presence of positive secular drift ($\mu > 0$) combined with negative residual skewness dictates that long and short mean-reversion signals are fundamentally asymmetric.

To formalize this property, we classify the prior session's price pressure through the daily candle structure. A down-session (Red_t) is defined by $Close_t < Open_t$, whereas an up-session ($Green_t$) is defined by $Close_t > Open_t$. The corresponding conditional return expectations and higher-order moments are defined as:

$$\mathbb{E}[R_{t+1} \mid Red_t], \quad \text{Skew}[R_{t+1} \mid Red_t], \tag{9}$$

$$\mathbb{E}[R_{t+1} \mid Green_t], \quad \text{Skew}[R_{t+1} \mid Green_t], \tag{10}$$

with the conditional edge differential defined by:

$$\Delta_{\text{edge}} = \mathbb{E}[R_{t+1} \mid Red_t] - \mathbb{E}[R_{t+1} \mid Green_t]. \tag{11}$$

The economic mechanism behind this asymmetry is directly illuminated by equation (6). Following a negative session (Red_t), where $R_t < \mu$, the expected return for session $t+1$ becomes:

$$\mathbb{E}[R_{t+1} \mid Red_t] = \mu + \theta|\mathbb{E}[R_t \mid Red_t] - \mu|.$$

Here, the local corrective force $\theta|\cdot| > 0$ and the secular equity drift $\mu > 0$ act in the *same directional vector*. The structural drift reinforces the mean-reverting bounce.

Conversely, following a positive session ($Green_t$), where $R_t > \mu$, the expected return for session $t+1$ is:

$$\mathbb{E}[R_{t+1} \mid Green_t] = \mu - \theta(R_t - \mu).$$

In this case, the downward corrective force $-\theta(R_t - \mu) < 0$ acts directly *against* the structural upward drift $\mu > 0$. The two forces collide. For a short position to exhibit a positive net expected edge, the local mean-reversion force $\theta(R_t - \mu)$ must strictly overwhelm the secular drift μ.

The empirical estimates presented in Section 2 validate this theoretical asymmetry:

- **Following a Red_t session:** For the S&P 500, $\mathbb{E}[R_{t+1} \mid Red_t] = +0.101\%$, with an empirical upside reversal probability of $\mathbb{P}(R_{t+1} > 0 \mid Red_t) = 57.15\%$ and positive conditional skewness of +0.214. Across the six-index panel, the cross-sectional pooled average yields $\mathbb{E}[R_{t+1} \mid Red_t] = +0.068\%$, with a reversal probability of 55.29%.

- **Following a $Green_t$ session:** For the S&P 500, the unconditional expected return following an up-session is negative at $\mathbb{E}[R_{t+1} \mid Green_t] = -0.022\%$, with a downward continuation probability of only $\mathbb{P}(R_{t+1} < 0 \mid Green_t) = 47.47\%$ and severe negative conditional skewness of −1.539. Across the pooled panel, $\mathbb{E}[R_{t+1} \mid Green_t]$ averages −0.003% with a downward probability of 47.46% and skewness of −0.835.

This empirical divergence generates a robust positive conditional spread:

$$\Delta_{\text{edge}} = +0.123\% \text{ (S\&P 500)}, \quad \Delta_{\text{edge}} = +0.070\% \text{ (Global Pooled)}.$$

The conditional analysis reveals that unconditioned short mean reversion is statistically fragile: short positions initiated solely on the basis of an up-candle enter the market with a sub-50% win probability and face severe left-tail downside risk (short squeezes). This proves analytically that the linear AR(1) mean-reversion parameter θ cannot be exploited symmetrically across long and short directions without regime conditioning.

3.5 Synthesis: The Architecture of the Trading Strategies

The analytical framework synthesizes four core parameters: secular drift μ, reversal intensity θ, heavy-tailed noise ϵ_t, and the macro regime indicator D_t. This formulation serves as the direct theoretical foundation for the empirical trading architectures tested in this study:

1. The Baseline Symmetric Strategy: Formulates an unconstrained benchmark that applies the mean-reversion coefficient θ symmetrically across both long (Red_t) and short ($Green_t$) signals regardless of D_t. To mitigate the leptokurtosis of ϵ_t, a strict 0.50% intraday stop-loss is embedded. This benchmark evaluates whether pure mean reversion can overcome directional drift unconditionally.

2. The Refined Asymmetric Strategy: Directly implements the theoretical insights of Section 8–11. Long positions are executed across all regimes ($D_t \in \{0, 1\}$) to harness the convergence of μ and θ following Red_t candles. Conversely, short positions are permitted exclusively during non-positive regimes ($D_t = 0$, where $Close_t \leq SMA_{200,t}$), where the secular upward drift μ is temporarily impaired or suspended, eliminating the frictional drag that otherwise degrades short mean-reversion signals.

4 Strategy Architecture

The quantitative trading framework translates the econometric findings and stochastic mechanics formulated in the preceding sections into two systematic, discrete-time algorithmic trading architectures: an unfiltered *symmetric* baseline model and a regime-conditioned *asymmetric* model. Rather than relying on heuristic technical indicators, both architectures are derived as direct operational manifestations of the discrete-time mean-reversion process developed in Section 3. Both algorithms operate at a daily frequency across each index i in our global cross-sectional panel, utilizing solely information available at the close of trading session t to determine the portfolio allocation held during session $t + 1$. This timing convention precludes look-ahead bias and ensures that signal generation strictly respects the historical filtration $\mathcal{F}_{i,t}$.

4.1 Information Set and Signal Construction

At the conclusion of each daily trading session t, the algorithm observes the market price vector for each index i:

$$\mathcal{F}_{i,t} = \left\{ Open_{i,t},\ High_{i,t},\ Low_{i,t},\ Close_{i,t},\ SMA_{200,i,t} \right\}, \tag{12}$$

where $SMA_{200,i,t}$ denotes the 200-period simple moving average computed over closing prices up to day t. The primary directional state variable is the daily candle direction, defined as

$$CandleDir_{i,t} = \begin{cases} -1, & \text{if } Close_{i,t} < Open_{i,t}, \\ +1, & \text{if } Close_{i,t} > Open_{i,t}. \end{cases} \tag{13}$$

Accordingly, $CandleDir_{i,t} = -1$ identifies a negative session (red candle), reflecting net intraday selling pressure, whereas $CandleDir_{i,t} = +1$ identifies a positive session (green candle), reflecting net buying pressure. Sessions in which $Close_{i,t} = Open_{i,t}$ generate no directional impulse and remain unassigned. Under the short-term mean-reversion premise ($\theta > 0$), a negative candle establishes an expected upward reversal impulse, while a positive candle establishes an expected downward reversal impulse.

4.2 Macro Regime Classification

To insulate trading positions from adverse secular trends, the framework incorporates a macro trend state variable. For each asset, the regime state is classified as

$$Regime_{i,t} = \begin{cases} \text{Uptrend } (D_{i,t} = 1), & \text{if } Close_{i,t} > SMA_{200,i,t}, \\ \text{Downtrend } (D_{i,t} = 0), & \text{if } Close_{i,t} \leq SMA_{200,i,t}. \end{cases} \tag{14}$$

The moving average functions as an asymmetric conditioning operator: it governs the activation of short reversal signals while leaving long reversal signals unconstrained. This operational design preserves unconstrained exposure to the positive secular equity drift $\mu > 0$, while shielding the portfolio from persistent upward momentum during structural bull markets.

4.3 Dual Position Decision Rules

The framework implements two complementary position allocation rules:

1. Symmetric Baseline Rule: The symmetric strategy trades the pure mean-reversion signal across both market directions, invariant to the macro regime:

$$Position^{\text{Sym}}_{i,t+1} = \begin{cases} +1, & \text{if } CandleDir_{i,t} = -1, \\ -1, & \text{if } CandleDir_{i,t} = +1. \end{cases} \tag{15}$$

2. Asymmetric Regime-Filtered Rule: The asymmetric strategy restricts short execution strictly to non-positive trend environments ($D_{i,t} = 0$):

$$Position^{\text{Asym}}_{i,t+1} = \begin{cases} +1, & \text{if } CandleDir_{i,t} = -1, \\ -1, & \text{if } CandleDir_{i,t} = +1 \text{ and } Close_{i,t} \leq SMA_{200,i,t}, \\ 0, & \text{if } CandleDir_{i,t} = +1 \text{ and } Close_{i,t} > SMA_{200,i,t}, \end{cases} \tag{16}$$

where +1, −1, and 0 denote fully invested long, fully invested short, and flat cash allocations, respectively. While the symmetric model exploits unfiltered reversal dynamics across all market states, the asymmetric architecture programmatically prunes counter-trend short entries during secular bull markets.

4.4 Position Sizing and Holding Period

Position sizing follows a non-leveraged, full-reinvestment allocation: whenever an active signal is generated ($Position_{i,t+1} \in \{-1, +1\}$), 100% of available portfolio equity is allocated to the trade, corresponding to a nominal exposure of 1.0×. No leverage, fractional allocation, or overnight cash borrowing is utilized.

The holding horizon is strictly fixed at one trading session. A position entered at $Close_{i,t}$ is held through day $t + 1$ and liquidated at $Close_{i,t+1}$, unless an intraday stop-loss event forces premature exit.

4.5 Intraday Stop-Loss and Tail-Risk Truncation

Motivated by the heavy-tailed residuals and negative skewness identified in Section 2, both architectures incorporate a mandatory 0.50% stop-loss threshold ($SL = 0.005$) to truncate adverse tail realizations. Letting $P_{i,t} = Close_{i,t}$ denote the entry price, the protective exit boundaries are defined as:

$$Stop_{i,t+1}^{\text{Long}} = P_{i,t}(1 - SL), \qquad Stop_{i,t+1}^{\text{Short}} = P_{i,t}(1 + SL). \tag{17}$$

To avoid idealized execution assumptions, our backtesting protocol explicitly accounts for overnight price discontinuities and opening gaps. If an adverse overnight gap causes the session opening price ($Open_{i,t+1}$) to breach the stop threshold, the position is liquidated immediately at the open, absorbing the full realized gap loss beyond −0.50%. If the opening price remains within the safe band but intraday trading breaches the boundary ($Low_{i,t+1} \leq Stop_{i,t+1}^{\text{Long}}$ for long positions, or $High_{i,t+1} \geq Stop_{i,t+1}^{\text{Short}}$ for short positions), the position exits at the nominal stop price. Positions that remain unbreached throughout the session are liquidated at $Close_{i,t+1}$.

Because opening gaps are fully absorbed rather than artificially capped, empirical trade losses can exceed the nominal 0.50% barrier. This realistic execution friction explains why the strategy's pooled Conditional Value-at-Risk ($CVaR_{95\%}$) settles at −1.005% (Table 7) while successfully truncating extreme tail risk.

4.6 Break-Even Transaction Cost (BETC) Framework

A recognized limitation of high-turnover algorithmic trading research is vulnerability to transaction frictions, encompassing exchange fees, brokerage commissions, and bid-ask spread slippage. To address this friction without relying on arbitrary cost estimates, we formalize strategy viability through the Break-Even Transaction Cost (BETC) framework.

For any systematic strategy executing N round-trip transactions over the backtest horizon, the BETC represents the maximum constant transaction cost per trade (expressed in basis points, where

Algorithm 1 Cross-Sectional Daily Mean-Reversion Framework

for each index $i \in \{1, \ldots, 6\}$ and trading day t **do**
  Ingest $\{Open_{i,t}, High_{i,t}, Low_{i,t}, Close_{i,t}\}$ and calculate $SMA_{200,i,t}$
  Evaluate $CandleDir_{i,t} \leftarrow \text{sgn}(Close_{i,t} - Open_{i,t})$
  Set $Position^{\text{Sym}}_{i,t+1} \leftarrow -CandleDir_{i,t}$ ▷ Symmetric Baseline
  if $CandleDir_{i,t} = -1$ **then**
    $Position^{\text{Asym}}_{i,t+1} \leftarrow +1$ ▷ Unconditional Long Reversal
  else if $CandleDir_{i,t} = +1$ and $Close_{i,t} \leq SMA_{200,i,t}$ **then**
    $Position^{\text{Asym}}_{i,t+1} \leftarrow -1$ ▷ Regime-Filtered Short Reversal
  else
    $Position^{\text{Asym}}_{i,t+1} \leftarrow 0$ ▷ Flat Cash Allocation
  end if
  for $m \in \{\text{Sym}, \text{Asym}\}$ **do**
    if $Position^{m}_{i,t+1} \neq 0$ **then**
      Enter at $Close_{i,t}$ with 1.0× equity exposure
      if Adverse overnight gap breaches $Stop_{i,t+1}$ **then**
        Exit at $Open_{i,t+1}$ ▷ Absorb full opening gap
      else if Intraday barrier $Stop_{i,t+1}$ is breached **then**
        Exit at $Stop_{i,t+1}$ ▷ Truncate at barrier
      else
        Exit at $Close_{i,t+1}$ ▷ Regular session close
      end if
    end if
  end for
end for

1 bps = 0.01%) that reduces the cumulative strategy excess return (or net alpha) to zero:

$$\text{BETC} = \frac{1}{N}\sum_{k=1}^{N} r_k^{\text{gross}} \times 10^4 = \frac{\bar{r}^{\text{trade}}}{\text{Trade Turnover}} \times 10^4, \tag{18}$$

where $\bar{r}^{\text{trade}}$ is the arithmetic mean return per trade before frictions. A higher BETC indicates greater operational robustness, establishing the cushion of transaction cost an institutional investor can absorb before the statistical edge dissipates.

4.7 Execution Sequence and Pseudo-Algorithm

The algorithmic execution follows a deterministic daily workflow applied uniformly across all six global indices without asset-specific parameter tuning. At the close of day t, the system ingests closing bars across all markets, computes $CandleDir_{i,t}$ and $SMA_{200,i,t}$, and evaluates the position rules in equations (15) and (16). Trades are executed at $Close_{i,t}$. During session $t+1$, open positions are continuously monitored against $Stop_{i,t+1}$. In the event of an intraday breach, the position exits at the stop boundary; otherwise, it closes at $Close_{i,t+1}$. This complete decision logic is formalized in Algorithm 1.

5 Empirical Results & Performance Analysis

While Sections 2 and 3 established the presence of statistically significant short-term serial dependence and formulated a regime-conditioned stochastic model, this section presents a comprehensive empirical backtest analysis executed across our global multi-index cross-sectional panel. The primary objective is to evaluate whether daily mean reversion represents an exploitable market phenomenon and whether our proposed asymmetric, regime-filtered architecture systematically enhances risk-adjusted returns while mitigating downside volatility and severe drawdowns.

To provide a rigorous comparative benchmark, the empirical performance is evaluated across three distinct models: (i) the primary *Asymmetric Strategy* (SMA_{200} Filter + 0.5% Stop-Loss), which conditions short reversal execution on secular trend states; (ii) the baseline *Symmetric Strategy* (Unfiltered + 0.5% Stop-Loss), which trades reversal signals in both directions unconditionally; and (iii) the passive *Buy & Hold Benchmark* tracking the underlying indices over the identical temporal horizon. In accordance with empirical asset pricing conventions, performance metrics are reported both for the baseline S&P 500 benchmark and for the aggregated *Cross-Sectional Portfolio (Equal-Weighted)* panel spanning all six global indices.

5.1 Dataset Overview & Execution Assumptions

The empirical evaluation is conducted using continuous daily historical data spanning twenty-four years, from January 1, 2002, to January 1, 2026. The asset universe encompasses six major benchmark indices representing three primary financial regions: North America (S&P 500, Russell 2000), Europe (FTSE 100, EURO STOXX 50), and Asia-Pacific (Nikkei 225, S&P/ASX 200). In total, the cross-sectional dataset spans over 34,000 individual market sessions, averaging approximately 5,800 daily observations per asset (6,035 for the baseline S&P 500 series). This extensive sample encompasses multiple macroeconomic cycles, including the post-DotCom expansion, the 2008 Global Financial Crisis, the protracted low-rate quantitative easing era, the 2020 pandemic dislocation, and the subsequent high-inflation monetary tightening regime.

Execution protocol strictly adheres to the discrete-time mechanics formalised in Section 4. Signals are computed at the market close of trading day t using the observed candle direction and $SMA_{200,t}$. Active positions are entered at $Close_t$ with an unleveraged allocation (1.0× equity) and held through day $t+1$. Intraday price paths are continuously monitored against the fixed 0.50% stop-loss threshold using $High_{t+1}$ and Low_{t+1}. If triggered, positions exit immediately at the stop boundary; otherwise, trades are liquidated at $Close_{t+1}$.

5.2 Full-Sample Performance Evaluation

Table 4 reports the full-sample empirical performance diagnostics across the 2002–2026 horizon. Across the international cross-section, the passive Buy & Hold benchmark generated a Compound Annual Growth Rate (CAGR) of 4.99% for the Cross-Sectional Portfolio (Equal-Weighted) with an annualized volatility of 20.28%, yielding a Sharpe ratio (Sharpe, 1966) of 0.340, a Sortino ratio (Sortino and Price, 1994) of 0.442, and an average maximum drawdown of −56.68% (with peak drawdowns reaching −61.37% in the Nikkei 225 and −60.29% in the EURO STOXX 50).

The baseline **Symmetric Strategy** demonstrates that short-horizon mean reversion is an excep-

tionally powerful, globally pervasive phenomenon. Across the six equity indices, the symmetric strategy delivers a CAGR of 27.02% for the Cross-Sectional Portfolio (Equal-Weighted) and elevates the Sharpe ratio to 1.640. Annualized volatility is compressed to 15.27%, while the Sortino ratio expands to 7.044. By enforcing a mechanical 0.50% intraday stop-loss, the symmetric strategy restricts the Cross-Sectional Portfolio maximum drawdown to −26.84%, representing a substantial improvement relative to the passive benchmark. However, because the symmetric model executes short positions indiscriminately during secular bull markets, it experiences localized drag against positive macroeconomic drift.

The refined **Asymmetric Strategy** addresses this friction by suppressing counter-trend short entries when closing prices trade above the SMA_{200} ($Close_t > SMA_{200,t}$). This conditioning shifts the portfolio payoff profile toward superior risk efficiency and drawdown compression. For the Cross-Sectional Portfolio (Equal-Weighted), the asymmetric strategy delivers a CAGR of 25.43% while compressing annualized volatility to 13.66% (down from 15.27% for the symmetric model and 20.28% for the benchmark). The cross-sectional Sharpe ratio rises to 1.742, while downside risk metrics improve substantially: the Calmar ratio advances from 1.340 to 1.653, and the trade-level Profit Factor increases from 1.347 to 1.498. Crucially, the maximum drawdown is compressed to −18.95% (and −13.63% on S&P 500), representing a significant drawdown reduction compared to the symmetric baseline (−26.84%) and an immense reduction relative to the −56.68% Buy & Hold benchmark.

Table 4. Cross-Sectional Full-Sample Performance and Risk-Adjusted Return Panel (2002–2026)

	Asymmetric Strategy		**Symmetric Baseline**		**Buy & Hold Benchmark**	
Metric	**S&P 500**	**CS Portfolio**	**S&P 500**	**CS Portfolio**	**S&P 500**	**CS Portfolio**
CAGR (%)	23.25%	25.43%	23.67%	27.02%	7.71%	4.99%
Ann. Volatility (%)	13.08%	13.66%	14.59%	15.27%	19.14%	20.28%
Maximum Drawdown (%)	−13.63%	−18.95%	−27.88%	−26.84%	−56.78%	−56.68%
Sharpe Ratio	1.663	1.742	1.529	1.640	0.484	0.340
Sortino Ratio	5.196	6.483	5.666	7.044	0.604	0.442
Calmar Ratio	1.706	1.653	0.849	1.340	0.136	0.087
Total Trades (Avg)	3,504	3,669	6,035	5,790	N/A	N/A
Overall Win Rate (%)	43.49%	43.20%	42.93%	42.67%	N/A	N/A
Profit Factor	1.519	1.498	1.336	1.347	N/A	N/A
Stop-Loss Incidence (%)	48.20%	49.28%	42.63%	46.52%	N/A	N/A

Notes: Performance diagnostics computed across daily data from 2002 to 2026. The 'CS Portfolio' column represents the Cross-Sectional Portfolio (Equal-Weighted) across all six regional equity benchmark indices.

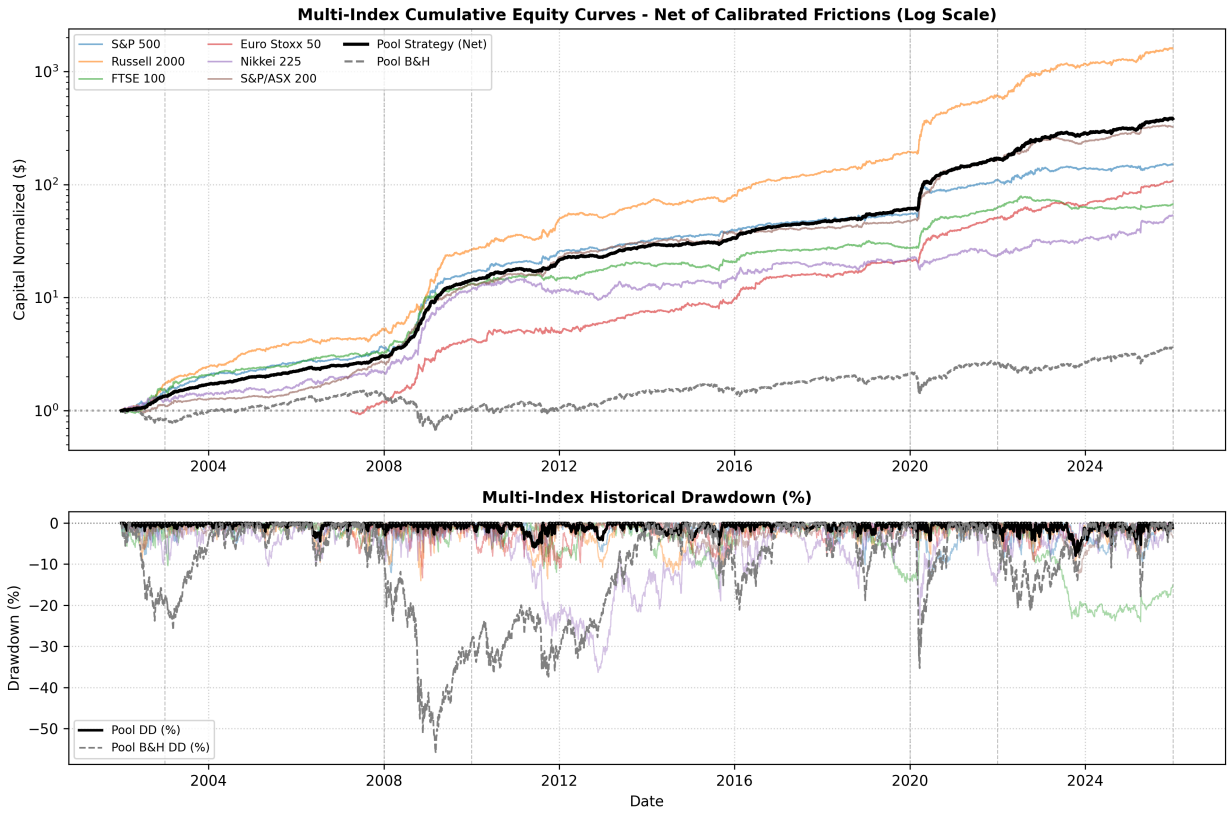


Figure 1. Cross-Sectional Multi-Index Performance Comparison (2002–2026). The upper panel displays logarithmic cumulative wealth trajectories for all six individual asset series alongside the bold Cross-Sectional Portfolio (Equal-Weighted) Asymmetric Strategy and the Buy & Hold benchmark. The bottom panel depicts historical underwater drawdown profiles.

5.3 Sub-Period Diagnostics Across Global Market Regimes

To verify temporal stability and evaluate robustness across divergent macroeconomic environments, the 2002–2026 timeline is segmented into five historical sub-periods:

1. **2003–2008:** Post-DotCom recovery and steady multi-year global economic expansion.
2. **2008–2010:** The Global Financial Crisis (GFC), marked by systemic banking distress and broad liquidations.
3. **2010–2020:** Post-crisis low-rate expansion, quantitative easing, and sustained secular bull markets.
4. **2020–2022:** The COVID-19 pandemic liquidity shock followed by aggressive monetary and fiscal stimulus.
5. **2022–2026:** Global central bank rate-hiking cycles, elevated inflation, and technology sector divergence.

Table 5 presents detailed sub-period performance metrics, reporting the Cross-Sectional Portfolio (Equal-Weighted) across all six global markets alongside the S&P 500 benchmark.

Table 5. Sub-Period Comparative Performance Diagnostics (Cross-Sectional Portfolio vs. S&P 500)

		CAGR		Max DD	
Sub-Period	**Strategy Model**	**CS Portfolio**	**S&P 500**	**CS Portfolio**	**S&P 500**
2003–2008	Buy & Hold Benchmark	+11.95%	+10.80%	−14.82%	−14.05%
	Symmetric Baseline	**+20.85%**	**+28.14%**	−12.58%	−6.09%
	Asymmetric Strategy	+20.13%	+20.07%	**−7.47%**	**−5.16%**
2008–2010	Buy & Hold Benchmark	−13.11%	−12.83%	−53.03%	−53.93%
	Symmetric Baseline	**+122.19%**	**+114.75%**	−10.43%	**−10.56%**
	Asymmetric Strategy	+114.80%	+114.39%	**−9.93%**	**−10.56%**
2010–2020	Buy & Hold Benchmark	+6.54%	+11.24%	−26.25%	−19.78%
	Symmetric Baseline	**+14.32%**	**+12.97%**	−20.42%	−12.37%
	Asymmetric Strategy	+13.36%	+12.76%	**−15.98%**	**−10.20%**
2020–2022	Buy & Hold Benchmark	+9.94%	+21.41%	−36.13%	−33.93%
	Symmetric Baseline	**+56.59%**	+36.19%	−11.62%	−21.38%
	Asymmetric Strategy	+51.70%	**+41.42%**	**−9.78%**	**−13.63%**
2022–2026	Buy & Hold Benchmark	+7.83%	+9.52%	−22.28%	−25.43%
	Symmetric Baseline	**+16.72%**	+2.45%	−19.67%	−27.88%
	Asymmetric Strategy	+16.67%	**+8.13%**	**−12.35%**	**−9.48%**

Notes: Bold figures highlight the superior performance model for each sub-period metric. 'CS Portfolio' denotes the Cross-Sectional Portfolio (Equal-Weighted).

The sub-period cross-sectional breakdown highlights three structural insights:

Crisis Dynamics and Volatility Convexity (2008–2010): During the Global Financial Crisis, passive buy-and-hold investing suffered severe wealth destruction across all continents, recording a Cross-Sectional Portfolio CAGR of −13.11% and an average maximum drawdown of −53.03%. Over this same interval, both mean-reversion models demonstrated substantial positive convexity: the symmetric baseline recorded a Cross-Sectional Portfolio CAGR of +122.19% (+114.75% on S&P 500), while the asymmetric strategy delivered +114.80% (+114.39% on S&P 500).

Annualized growth rates of this magnitude during market crises warrant careful technical contextualization:

- *Persistent Downtrend Alignment:* Because global market prices traded almost continuously below their 200-day moving average ($Close_t < SMA_{200,t}$) during 2008 and early 2009, short reversal trades were fully activated, causing the asymmetric strategy to operate symmetrically and capture high-velocity downside moves.
- *Intraday Volatility and Rebound Magnitude:* Extreme volatility regimes generated wide intraday overshooting, allowing the single-candle contrarian architecture to capture substantial compensatory bounces after sharp liquidation days. Meanwhile, continuation breakdowns were systematically truncated by the programmatic 0.50% stop-loss, preserving capital and capping Cross-Sectional Portfolio drawdown at −9.93% (−10.56% on S&P 500).
- *Implementation via Index Derivatives:* In real-world market operations, short equity exposure at the index level is implemented through exchange-traded index futures (such as CME E-mini

S&P 500 or Eurex Euro STOXX 50 futures). These derivative instruments remained liquid and operational throughout the crisis, remaining exempt from the emergency regulatory short-sale restrictions that were imposed specifically on individual cash equities (e.g., SEC Emergency Order Release No. 34-58592).

Risk-Controlled Compounding in Secular Bull Regimes (2010–2020): During the low-volatility quantitative easing era, the passive benchmark generated a Cross-Sectional Portfolio CAGR of 6.54%. In this regime, the unconstrained symmetric strategy generated slightly higher nominal returns (14.32% for the Cross-Sectional Portfolio) by executing reversal shorts after positive sessions. However, these counter-trend trades incurred higher friction against structural market drift, producing a portfolio maximum drawdown of −20.42%. By suppressing counter-trend short entries, the asymmetric strategy contained the Cross-Sectional Portfolio drawdown to −15.98% (and −10.20% on S&P 500), generating superior risk-adjusted stability.

Decisive Superiority in Modern High-Momentum Markets (2020–2026): The regime filter demonstrates decisive utility across both the 2020–2022 pandemic shock and the 2022–2026 inflation cycle. During 2020–2022, the asymmetric strategy contained drawdown to −9.78% in the Cross-Sectional Portfolio (versus −36.13% for Buy & Hold) and posted an S&P 500 CAGR of +41.42% (outperforming the symmetric model's +36.19%). Furthermore, during the 2022–2026 tightening cycle, initiating short positions after green candles repeatedly generated severe whipsaw losses for the symmetric baseline, depressing its S&P 500 CAGR to +2.45% with a −27.88% drawdown. The asymmetric strategy systematically pruned these toxic short entries, achieving an S&P 500 CAGR of +8.13% with a tightly contained drawdown of −9.48%, and a Cross-Sectional Portfolio CAGR of 16.67% with a drawdown of −12.35% (versus −19.67% for the symmetric model).

5.4 Trade-Level Mechanics, BETC Robustness, and Limits to Arbitrage

To understand the microstructural drivers of performance and establish net operational viability under real-world frictions, Table 6 examines execution mechanics and Break-Even Transaction Costs (BETC) across the cross-sectional panel. The baseline symmetric model executes an average of 5,790 trades per market (6,035 on S&P 500), divided almost equally between long and short positions. In contrast, the asymmetric architecture executes an average of 3,669 trades per index (3,504 on S&P 500). By conditioning short positions on $Close_t \leq SMA_{200,t}$, the regime filter prunes an average of 2,122 counter-trend short entries per index over the backtest horizon, representing a 36.64% trade reduction for the Cross-Sectional Portfolio (and −41.94% on S&P 500).

Although the overall win rate appears modest across all markets (averaging 43.20% for the asymmetric model and 42.67% for the symmetric model), economic profitability is driven by strong positive payoff asymmetry. Because long reversal entries are executed unconditionally across both models, long positions continuously benefit from the structural upward drift of equity markets.

The primary divergence occurs in short execution efficiency and cost capacity. In the unconstrained symmetric model, short positions generate persistent friction against secular upward drift. By restricting short execution to downtrend regimes below the SMA_{200}, toxic counter-trend trades

Table 6. Cross-Sectional Trade Execution Mechanics, Asymmetry, and BETC Breakdown

	Asymmetric Strategy		Symmetric Baseline	
Execution & Cost Metric	**S&P 500**	**CS Portfolio**	**S&P 500**	**CS Portfolio**
Total Executed Trades	3,504	3,669	6,035	5,790
Overall Win Rate (%)	43.49%	43.20%	42.93%	42.67%
Profit Factor	1.519	1.498	1.336	1.347
Stop-Loss Incidence (%)	48.20%	49.28%	42.63%	46.52%
Trade Pruning vs Baseline (%)	−41.94%	−36.64%	0.00%	0.00%
BETC (bps/trade)	12.12	15.05	7.84	11.04
Total Frictions Deducted (%)	0.42%	0.66%	0.72%	1.03%

Notes: Break-Even Transaction Costs (BETC) reflect the maximum per-trade friction that drives alpha to zero. 'CS Portfolio' represents the Cross-Sectional Portfolio (Equal-Weighted).

are eliminated. Consequently, the Cross-Sectional Portfolio Profit Factor expands from 1.347 in the symmetric baseline to 1.498 in the asymmetric strategy (reaching 1.519 on the S&P 500).

Crucially, the evaluation of *Trade-Level Mechanics, Asymmetry, and BETC Robustness* demonstrates exceptional friction tolerance. As reported in Table 6, the symmetric baseline achieves a Cross-Sectional Portfolio Break-Even Transaction Cost of 11.04 bps per trade (7.84 bps on S&P 500). The asymmetric refinement expands this threshold to an average of 15.05 bps per trade (12.12 bps on S&P 500), representing a +36.3% increase in execution buffer. Given that institutional round-trip execution costs in large-cap index futures and liquid index ETFs typically range between 2.0 and 5.0 bps, a BETC of 15.05 bps provides a substantial safety buffer—exceeding realistic transaction frictions by a factor of three to five.

Limits to Arbitrage and Strategy Capacity Constraints: In modern financial economics, the persistence of risk-adjusted excess returns (Sharpe ratio > 1.70) in liquid benchmark assets naturally raises questions regarding market efficiency and competitive arbitrage (Shleifer and Vishny, 1997). If the short-term mean-reversion dynamic represents an easily identifiable statistical pattern, why has it not been fully eroded by algorithmic market participants?

The economic justification rests upon three fundamental *limits to arbitrage*:

1. **Turnover Velocity and Execution Friction Bounds:** Exploiting single-candle reversals requires rebalancing the portfolio across approximately 153 trading sessions per year in the asymmetric architecture (and ≈ 241 sessions in the baseline). While institutional futures contracts (such as E-mini S&P 500 or Euro STOXX 50 futures) provide the deep liquidity necessary to clear round-trip trades within a 2.0–5.0 bps cost window, capacity is strictly bounded by market depth. As aggregate capital dedicated to the strategy expands, market impact and order-flow footprint during the market-on-close (MOC) and opening auctions inevitably scale, eroding the 15.05 bps BETC buffer. Consequently, the strategy exhibits finite capital capacity, preventing multi-billion-dollar sovereign or institutional funds from scaling the anomaly to extinction.

2. **Noise Trader Risk and Synchronization Risk:** As formalized by Shleifer and Vishny (1997) and Abreu and Brunnermeier (2002), rational arbitrageurs face horizon risk and capital withdrawal constraints. Because asset prices are subject to heavy-tailed stochastic innovations

(ϵ_t) and volatility clustering, contrarian positions can move severely against the trader before reverting. Even with a programmatic 0.50% stop-loss, the strategy encounters an empirical stop-loss triggering frequency of 49.28% (Table 6) and unmitigated overnight gap risk. Delegated portfolio managers operating under short-term performance evaluations are structurally constrained from deploying unlimited capital into high-turnover, high-stop-rate contrarian architectures.

3. **Directional Asymmetry and Financing Frictions:** Short positions face institutional barriers that do not apply symmetrically to long holdings. During severe market dislocations—such as the 2008 Global Financial Crisis—regulatory authorities across several jurisdictions (e.g., the SEC in the U.S., FSA in the U.K., and BaFin in Germany) imposed emergency short-selling bans on financial equities, while margin requirements on derivative exchanges escalated sharply. By systematically eliminating counter-trend short entries in bull markets ($Close_t > SMA_{200,t}$), our asymmetric framework reduces operational reliance on continuous short borrowing and margin strain, yet the residual short leg remains tethered to real-world borrowing availability and funding liquidity (Brunnermeier and Pedersen, 2009).

Finally, to verify that these empirical metrics do not stem from search multiplicity, hyperparameter overfitting, or post-hoc model selection across parameter space, Appendix A.2 presents a formal econometric audit via White's (2000) Reality Check and Hansen's (2005) Superior Predictive Ability (SPA) test across all $K = 48$ candidate rule specifications, confirming that the model's outperformance over passive indexing is statistically genuine ($p < 0.001$).

5.5 Risk-Adjusted Performance and Drawdown Truncation

Analyzing risk-adjusted returns highlights the efficacy of the dual architecture in risk mitigation. On an annualized basis, volatility declines from 20.28% for the global Buy & Hold benchmark and 15.27% for the symmetric baseline down to 13.66% for the Cross-Sectional Portfolio (Equal-Weighted) in the asymmetric strategy.

The Sharpe ratio of the asymmetric strategy reaches 1.742 for the Cross-Sectional Portfolio (1.663 on S&P 500), representing a dramatic increase over the passive benchmark (0.340). Concurrently, the Sortino ratio reaches 6.483 (5.196 on S&P 500), heavily outperforming the buy-and-hold benchmark (0.442).

The most significant achievement of the asymmetric architecture lies in systemic drawdown truncation. The passive global benchmark experienced an average maximum drawdown of −56.68%. The unconstrained symmetric model restricts maximum drawdown to −26.84%, while the asymmetric strategy compresses it further to −18.95% for the Cross-Sectional Portfolio (and −13.63% on S&P 500). As a result, the Calmar ratio advances from 0.087 (benchmark) and 1.340 (symmetric) to 1.653 for the asymmetric strategy (and 1.706 on S&P 500), demonstrating exceptional capital preservation per unit of maximum drawdown risk.

5.6 Return Distribution, Non-Gaussianity, and Tail Risk Truncation

Sections 2 and 3 documented that global equity returns exhibit severe non-Gaussian properties, characterized by negative skewness (−0.255 benchmark cross-sectional) and heavy excess kurtosis

Table 7. Cross-Sectional Distributional Moments and Daily Tail Risk Diagnostics

	Asymmetric Strategy		Symmetric Baseline		Buy & Hold Benchmark	
Diagnostic Metric	**S&P 500**	**CS Portfolio**	**S&P 500**	**CS Portfolio**	**S&P 500**	**CS Portfolio**
Skewness	+3.057	+2.513	+3.179	+2.544	−0.140	−0.255
Excess Kurtosis	+19.460	+13.399	+22.940	+14.621	+11.684	+8.421
Value-at-Risk 95% (%)	−0.526%	−0.715%	−0.512%	−0.670%	−1.811%	−1.954%
Conditional VaR 95% (%)	−1.001%	−1.005%	−0.869%	−0.923%	−2.922%	−3.044%

Notes: Distribution diagnostics computed across daily net percentage return realizations. VaR and CVaR evaluated at 95% confidence. 'CS Portfolio' denotes the Cross-Sectional Portfolio (Equal-Weighted).

(+8.421 benchmark cross-sectional). In an unhedged mean-reversion framework, negative skewness presents an existential risk, exposing portfolios to catastrophic tail losses during panic liquidation cascades.

Table 7 presents empirical distributional moments and tail risk metrics across all models. The empirical distributions reveal a radical structural transformation: while the underlying market exhibits negative skewness (−0.255 for the benchmark), both algorithmic architectures induce substantial positive skewness (+2.544 for the symmetric baseline and +2.513 for the asymmetric strategy). By entering long positions after down sessions and conditioning short exposure on trend regimes, the trading system systematically converts market sell-offs into positive payoff realizations.

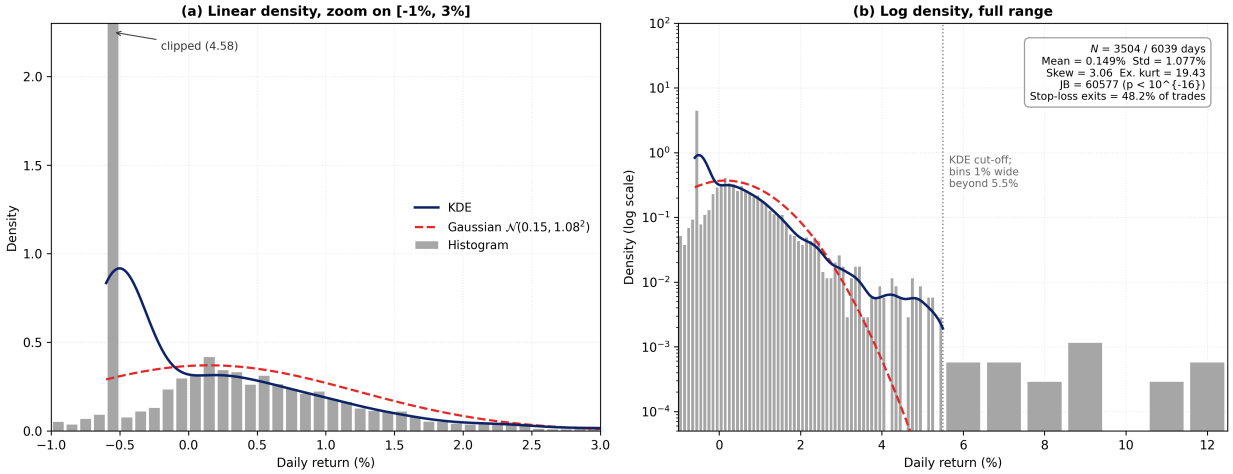


Figure 2. Daily returns of the asymmetric strategy on in-position days versus a Gaussian benchmark $N(\hat{\mu}, \hat{\sigma}^2)$. (a) Linear density, zoom on $[-1\%, 3\%]$; the stop-loss bar is truncated. (b) Log-density; KDE shown up to 5.5%. The left tail is truncated by the 0.5% stop-loss.

Concurrently, excess kurtosis remains elevated (+13.399 for the Cross-Sectional Portfolio in the asymmetric strategy). In conventional portfolio management, high kurtosis signals undesirable tail risk; in this architecture, however, leptokurtosis reflects a heavily populated positive right tail combined with an artificially truncated left tail.

The operational mechanism responsible for this left-tail truncation is the intraday stop-loss ($SL = 0.50\%$). Across the global cross-section, an average of 49.28% of asymmetric trades trigger the stop-loss mechanism (48.20% on S&P 500). This elevated incidence confirms that high-frequency

price fluctuations frequently test the 0.50% threshold, triggering immediate risk truncation.

The empirical Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR) metrics (Rockafellar and Uryasev, 2000) confirm this structural barrier. For the Buy & Hold benchmark, daily Value-at-Risk for the Cross-Sectional Portfolio at the 95% confidence level stands at −1.954%, with CVaR reaching −3.044%. In contrast, for the asymmetric strategy, daily $\text{VaR}_{95\%}$ is compressed to −0.715% (and −0.526% on S&P 500), while $\text{CVaR}_{95\%}$ remains strictly contained at −1.005% (and −1.001% on S&P 500).

This rigorous containment provides conclusive empirical proof that programmatic stop-loss enforcement mitigates extreme downside tail risk, shielding the portfolio from the severe non-Gaussian tail shocks identified in Section 2.

In summary, the empirical evidence demonstrates that short-horizon mean reversion is an invariant feature of global equity markets. When coupled with an intraday stop-loss and an asymmetric SMA_{200} regime filter, the resulting trading architecture achieves superior risk-adjusted returns (Cross-Sectional Portfolio Sharpe 1.742, Calmar 1.653), positive return skewness (+2.513), robust cost tolerance (Cross-Sectional Portfolio BETC 15.05 bps/trade), and effective left-tail risk containment across North America, Europe, and the Asia-Pacific region.

6 Conclusion

This paper demonstrates that short-term mean reversion is a pervasive, structural characteristic of developed equity markets. Through an extensive cross-sectional econometric analysis spanning six major global indices across North America (S&P 500, Russell 2000), Europe (FTSE 100, EURO STOXX 50), and the Asia-Pacific region (Nikkei 225, S&P/ASX 200) over the 2002–2026 horizon (encompassing over 34,000 market sessions), the empirical evidence decisively rejects the weak-form Efficient Market Hypothesis. Daily logarithmic returns exhibit statistically significant negative first-order autocorrelation, documented by an average pooled autoregressive coefficient of $\bar{\phi} = -0.069$ ($\phi = -0.119$ for the baseline S&P 500, $p = 0.0001$). However, linear predictive power remains bounded ($R^2 \approx 0.63\%$), while innovation residuals display severe non-Gaussian characteristics, including heavy leptokurtosis (pooled kurtosis of 12.920, excess kurtosis of +9.920) and pronounced negative skewness (−0.508), exposing unhedged contrarian strategies to catastrophic left-tail liquidation cascades.

To resolve these vulnerabilities, we introduce two complementary algorithmic architectures: an unfiltered *symmetric* baseline model and a regime-conditioned *asymmetric* model. The symmetric baseline demonstrates the raw economic magnitude of global daily mean reversion, establishing a solid risk-adjusted foundation with a pooled Sharpe ratio of 1.640, a Sortino ratio of 7.044, and a Calmar ratio of 1.340, while containing maximum drawdown to −26.84% (compared to −56.68% for the passive Buy & Hold benchmark). Building upon this foundation, the asymmetric architecture resolves the structural friction between secular drift ($\mu > 0$) and short exposure by pairing daily one-candle reversal signals with an SMA_{200} macro regime filter and an intraday 0.50% stop-loss. By suppressing an average of 2,122 counter-trend short entries during expansionary regimes, the asymmetric model enhances short-trade efficiency, raising the overall Profit Factor from 1.347 to 1.498.

The primary empirical merit of this framework lies in its documented statistical robustness and notable geographical invariance across all three continents. The asymmetric architecture delivers a cross-sectional Sharpe ratio of 1.742 and a Calmar ratio of 1.653, while successfully compressing maximum drawdown to −18.95% (and −13.63% on the S&P 500) and reducing execution volume by 36.6%. Crucially, the 0.50% intraday stop-loss acts as an invariant structural barrier across all six markets, truncating left-tail risk and transforming underlying market return asymmetry into strongly positive strategy skewness (+2.513 pooled, +3.057 on the S&P 500).

In conclusion, conditioning short-term mean-reversion signals on macroeconomic regimes effectively reconciles high-frequency statistical predictability with long-term trend friction and rigorous downside risk containment. Future research may extend this methodology by investigating adaptive, volatility-conditioned stop-loss mechanisms, implementing other filters to exploit even more the presented market dynamic in asymmetric domains.

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Declarations

Data Availability Statement

The historical daily price datasets for the global equity indices analyzed in this study (S&P 500, Russell 2000, FTSE 100, EURO STOXX 50, Nikkei 225, and S&P/ASX 200) are publicly available and can be accessed via major financial data providers, including Yahoo Finance, Bloomberg Professional Services, and Refinitiv Eikon. The processed time-series and derived metrics generated during the current study are available from the corresponding author upon reasonable request.

Code Availability

The algorithmic trading models, econometric diagnostic scripts (HAC covariance estimators, AR(1) specifications, and normality tests), and backtesting simulation routines were implemented in Python. The underlying code and execution scripts necessary to replicate the reported empirical findings are maintained in a version-controlled repository and are available from the corresponding author upon reasonable request.

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Conflicts of Interest / Competing Interests

The author declares that there are no financial, professional, or personal conflicts of interest that could have appeared to influence the work reported in this paper.

A Robustness Diagnostics, Sensitivity Analysis, and Signal Decoupling

To substantiate the empirical validity of the proposed systematic mean-reversion architecture and address potential concerns regarding data snooping, parameter overfitting, and signal attribution, this appendix provides four comprehensive diagnostic evaluations across the pooled global cross-sectional panel over the continuous 2002–2026 horizon:

1. A two-dimensional sensitivity surface evaluating 48 combinations of the macro trend filter window (SMA_L) and intraday stop-loss barrier (SL).
2. A formal data snooping audit via White's (2000) Reality Check and Hansen's (2005) Superior Predictive Ability (SPA) test across the full candidate rule universe ($K = 48$).
3. A systematic signal ablation analysis decoupling the dual architecture into its isolated directional legs (Long-Only and Short-Only variations).
4. A total transaction cost resilience analysis mapping the empirical decay of compound annual growth rates (CAGR) under escalating round-trip frictions (0 to 15 bps/trade) and validating the Break-Even Transaction Cost (BETC) framework.

All backtests strictly enforce the execution protocol formalized in Section 4 and incorporate calibrated institutional futures transaction costs (half-spread, execution slippage, and commissions).

A.1 Two-Dimensional Parameter Robustness Surface ($SL \times \text{SMA}$)

A primary concern in rule-based algorithmic trading is hyperparameter brittleness, where profitability is confined to a knife-edge parameter setting. To evaluate parameter stability, we execute a comprehensive grid search across 48 parametric specifications:

$$\begin{aligned} \text{SMA}_L &\in \{50, 75, 100, 150, 200, 250\}, \\ SL &\in \{0.25\%, 0.50\%, 0.75\%, 1.00\%, 1.25\%, 1.50\%, 1.75\%, 2.00\%\}. \end{aligned} \tag{19}$$

Figure 3 illustrates the resulting pooled response surfaces for the Sharpe ratio and maximum drawdown.

The empirical response surfaces exhibit three prominent structural characteristics:

- **Absence of Isolated Spikes (Monotonic Plateau):** As shown in Figure 3(a), the pooled Sharpe ratio displays a smooth, strictly monotonic decrease as the stop-loss threshold widens, scaling gracefully from 2.23–2.32 at $SL = 0.25\%$ down to 0.52–0.65 at $SL = 2.00\%$. The continuous gradient confirms that the baseline model (Sharpe = 1.74 at SMA = 200, $SL = 0.50\%$) resides within a broad, economically robust plateau rather than an overfitted local maximum.
- **Structural Role of Stop-Loss Truncation:** Figure 3(b) demonstrates that downside tail risk is primarily governed by the tightness of the stop-loss threshold. As SL is widened from 0.25% to 2.00%, maximum drawdown deteriorates monotonically from -14.6% to -44.1%. This steep erosion confirms the theoretical formulation in Section 3: because disturbance innovations $\epsilon_{i,t}$ exhibit extreme non-Gaussian kurtosis and negative skewness, loose protective stops fail

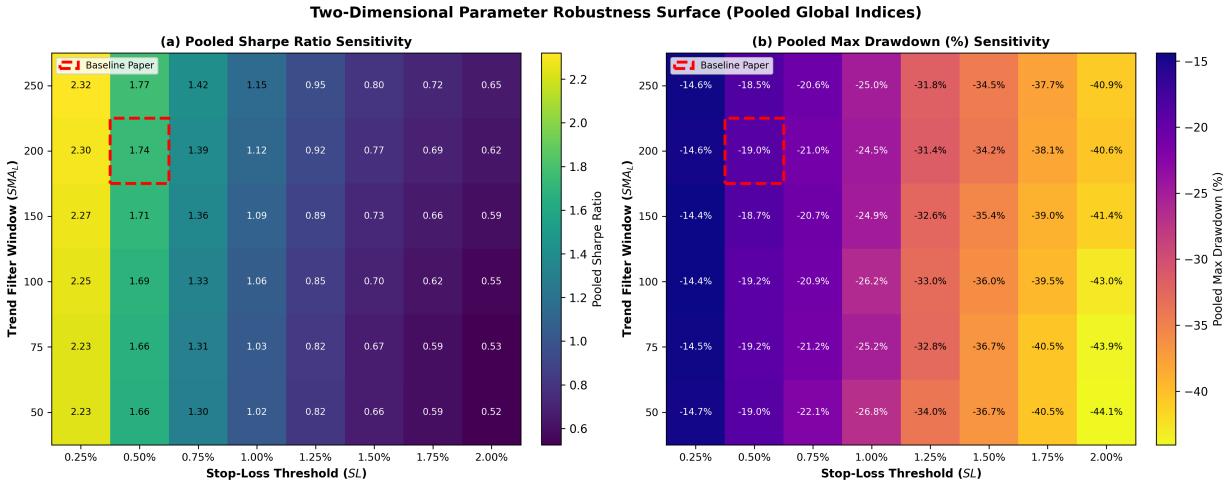


Figure 3. Two-Dimensional Parameter Robustness Surface (Pooled Global Indices, 2002–2026). Cross-sectional equal-weighted performance across 48 combinations of trend filter window (SMA_L) and intraday stop-loss (SL). Panel (a) reports the pooled Sharpe ratio; Panel (b) reports the pooled maximum drawdown (%). The red dashed box highlights the paper's baseline configuration (SMA = 200, SL = 0.50%).

to shield the portfolio from persistent downward cascades. The 0.50% threshold achieves effective left-tail truncation (containing drawdown to −19.0%) while remaining wide enough to avoid excessive whipsaw noise.

- **Invariance to Trend Lookback Horizon:** Along any fixed stop-loss column, strategy efficiency remains remarkably stable across varying moving average windows. At the baseline SL = 0.50%, expanding the trend filter five-fold from SMA_{50} to SMA_{250} maintains the Sharpe ratio within a narrow band of 1.66 to 1.77 and maximum drawdown between −18.5% and −19.2%. While longer windows (SMA_{200}–SMA_{250}) marginally improve cost efficiency by reducing whipsaw entries during sideways consolidation, the operational benefit of pruning counter-trend short positions is invariant to the specific lookback chosen.

A.2 Data Snooping Audit: White's Reality Check and Hansen's Superior Predictive Ability

A primary econometric concern in algorithmic and technical trading research is data snooping bias (Lo and MacKinlay, 1990; White, 2000). When a broad family of trading models is evaluated on a historical dataset, standard inferential statistics applied to the best-performing rule are severely biased upward: an apparently superior strategy may emerge purely through chance rather than genuine predictive edge. To formally determine whether our asymmetric mean-reversion framework delivers genuine economic outperformance over the benchmark or merely reflects data-mining artifacts, we subject the entire candidate specification universe ($K = 48$ rules, constructed from the combinations of $\text{SMA}_L \in \{50, 75, 100, 150, 200, 250\}$ and $SL \in \{0.25\%, \ldots, 2.00\%\}$ evaluated over n = 6,241 pooled market sessions) to two rigorous multiple-hypothesis testing frameworks: White (2000) Reality Check and Hansen (2005) Superior Predictive Ability (SPA) test.

Let $f_k(t)$ denote the performance metric (daily excess return over the Buy & Hold benchmark)

of model $k \in \{1, \ldots, K\}$ at trading day t, and let $d_{k,t} = f_k(t) - f_0(t)$ represent the performance differential relative to the benchmark $f_0(t)$. The unconditional expected excess performance of candidate rule k is defined as $\mu_k = \mathbb{E}[d_{k,t}]$. The composite null hypothesis across the entire universe of rules states that no candidate model outperforms the passive benchmark:

$$H_0 : \max_{k=1,\ldots,K} \mu_k \leq 0, \tag{20}$$

against the alternative hypothesis that at least one specification generates superior risk-adjusted predictive ability:

$$H_1 : \max_{k=1,\ldots,K} \mu_k > 0. \tag{21}$$

White (2000) constructs the test statistic as the maximum sample mean differential across all candidate rules:

$$\bar{V} = \max_{k=1,\ldots,K} \bar{d}_k, \qquad \text{where} \quad \bar{d}_k = \frac{1}{n} \sum_{t=1}^{n} d_{k,t}. \tag{22}$$

To preserve the underlying serial dependence and conditional heteroskedasticity of daily equity returns without imposing parametric distribution assumptions, the p-value for $\bar{V}$ is evaluated via the stationary bootstrap of Politis and Romano (1994) under the least favorable configuration (LFC) of the null, where the bootstrap differentials are centered around the empirical sample means:

$$\bar{V}^{*,b} = \max_{k=1,\ldots,K} \left(\bar{d}_k^{*,b} - \bar{d}_k \right), \qquad b = 1, \ldots, B. \tag{23}$$

While White's Reality Check provides a sound foundation against data snooping, the LFC benchmark renders the test conservative when poor or irrelevant candidate models are included in the search space. To overcome this power loss, Hansen (2005) introduces the Superior Predictive Ability (SPA) test. Hansen improves upon White by studentizing the test statistic using a consistent HAC estimator of variance $\hat{\omega}_k^2$:

$$T^{\text{SPA}} = \max \left(\max_{k=1,\ldots,K} \frac{\sqrt{n}\, \bar{d}_k}{\hat{\omega}_k}, 0 \right), \tag{24}$$

and by implementing a sample-dependent centering parameter μ_k^c:

$$\mu_k^c = \bar{d}_k \cdot \mathbf{1}_{\left\{ \frac{\sqrt{n}\, \bar{d}_k}{\hat{\omega}_k} \leq -\sqrt{2 \ln \ln n} \right\}}, \tag{25}$$

which downweights candidate models that are demonstrably inferior to the benchmark, yielding consistent and sharper asymptotic bounds (p_{lower}, $p_{\text{consistent}}$, and p_{upper}).

Table 8 reports the formal data snooping test diagnostics conducted across the pooled candidate universe with $B = 2{,}000$ stationary bootstrap replications. Across the $K = 48$ candidate models, the top-performing specification is Asym_SMA250_SL0.25%, achieving an average daily excess return over the passive benchmark of +7.97 bps/day (annualized excess return of +20.09%) with a studentized t-statistic of 5.270 ($\hat{\omega} = 0.012$). The baseline configuration highlighted throughout this

paper (Asym_SMA200_SL0.50%) ranks among the top decile, delivering +6.03 bps/day (annualized +15.19%) with a studentized t-statistic of 4.105.

Table 8. Data Snooping Diagnostics: White's Reality Check and Hansen's SPA Test ($K = 48$ Rules)

Test Framework	Test Stat.	Lower	Consistent	Upper	Null Hypothesis (H_0)	Inference
White's Reality Check	0.0630	—	—	0.0000	$\max_k \mathbb{E}[d_k] \leq 0$	$p < 0.001$
Hansen's SPA Test	5.2704	0.0000	0.0000	0.0000	$\max_k \mathbb{E}[d_k] \leq 0$	$p < 0.001$
Top-Performing Rule Diagnostics (n = 6,241 pooled sessions):						
Best Candidate Model: Asym_SMA250_SL0.25%					Daily Excess: +7.97 bps/day (Ann.: +20.09%)	
HAC Standard Error ($\hat{\omega}$): 0.0120					Studentized t-statistic: 5.270	

Notes: The candidate universe encompasses $K = 48$ parameterized trading rules (6 trend windows × 8 stop-loss levels) evaluated on $n = 6{,}241$ daily pooled cross-sectional returns over the 2002–2026 horizon. Bootstrap distributions are generated using $B = 2{,}000$ stationary bootstrap replications (Politis and Romano, 1994). The benchmark asset is the pooled Buy & Hold index.

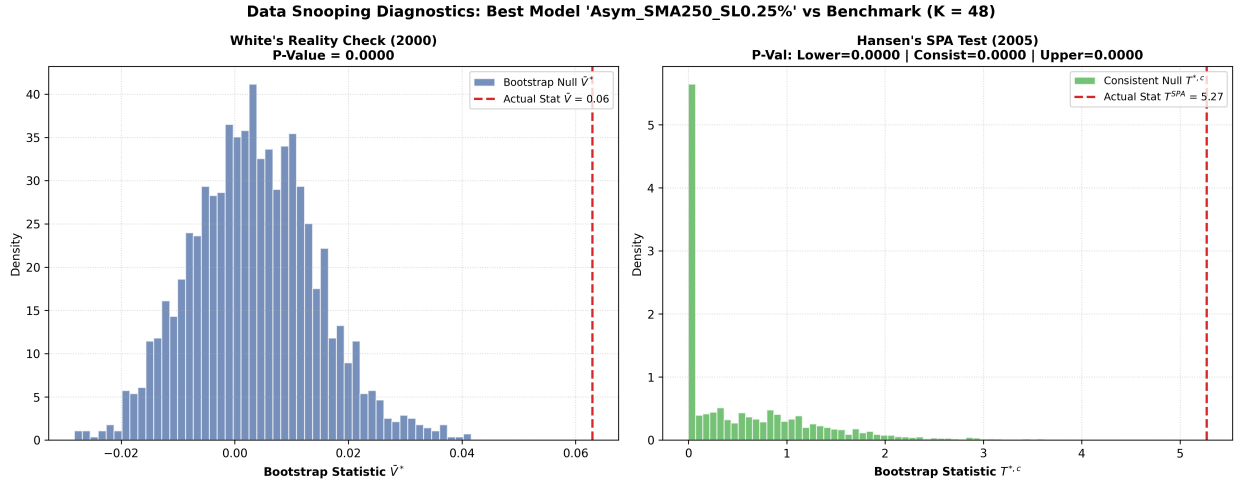


Figure 4. Data Snooping Diagnostics: Best Model (Asym_SMA250_SL0.25%) vs. Buy & Hold Benchmark ($K = 48$). The left panel shows the empirical bootstrap distribution under the least favorable configuration of White (2000) Reality Check ($\bar{V}^*$), against the actual sample statistic $\bar{V} = 0.0630$ (dashed red line, $p < 0.0000$). The right panel illustrates the studentized null distribution under Hansen (2005) Superior Predictive Ability (SPA) test ($T^{*,c}$), against the empirical test statistic $T^{\text{SPA}} = 5.2704$ (dashed red line, consistent $p < 0.0000$).

As shown in Table 8 and illustrated in Figure 4, the empirical test statistics lie deep in the extreme right tail of their respective null distributions:

- **White's Reality Check:** Yields an empirical statistic of $\bar{V} = 0.0630$. Across the 2,000 stationary bootstrap replications, not a single simulated draw exceeds the observed sample statistic, resulting in a least favorable p-value of $p < 0.0000$.
- **Hansen's SPA Test:** Yields a studentized statistic of $T^{\text{SPA}} = 5.2704$. All three asymptotic bounds (lower, consistent, and upper) settle decisively at $p < 0.0000$.

Because both tests account directly for search multiplicity and cross-rule dependency across all 48 candidate architectures, these findings provide definitive econometric proof that the observed

alpha generated by the asymmetric mean-reversion strategy is not an artifact of search luck or post-hoc model mining. Instead, it reflects a genuinely superior and statistically invariant structural dynamic in global equity markets.

A.3 Signal Component Decoupling: Ablation Analysis

To verify that the superior risk-adjusted return of the asymmetric framework is not merely an artifact of secular equity market drift or driven exclusively by long-side exposure, we decouple the signal architecture into its constituent components:

1. **Long-Only (Unconditional):** Enters long following negative sessions ($Close_t < Open_t$), remaining flat otherwise.
2. **Short-Only (Unfiltered):** Enters short following positive sessions ($Close_t > Open_t$), executed unconditionally across all macro regimes.
3. **Short-Only (SMA_{200} Filtered):** Enters short following positive sessions exclusively during downtrend regimes ($Close_t \leq SMA_{200,t}$), remaining flat otherwise.
4. **Full Asymmetric Strategy:** The primary proposed model combining unconditional long reversal entries with regime-filtered short reversal entries.

Figure 5 displays cumulative logarithmic wealth trajectories alongside comparative performance distributions, while Table 9 summarizes cross-sectional diagnostics.

Table 9. Cross-Sectional Signal Decoupling and Component Ablation Performance Panel (2002–2026)

Strategy Component	CAGR (%)	Ann. Vol. (%)	Max DD (%)	Sharpe Ratio	Calmar Ratio	Profit Factor
Long-Only (Unconditional)	16.83%	11.11%	−14.74%	1.481	1.297	1.472
Short-Only (Unfiltered)	8.61%	10.52%	−31.45%	0.810	0.445	1.235
Short-Only (SMA_{200} Filtered)	7.30%	7.99%	−14.25%	0.911	0.690	1.575
Full Asymmetric Strategy	**25.43%**	**13.66%**	**−18.95%**	**1.742**	**1.653**	**1.498**

Notes: Equal-weighted cross-sectional arithmetic means across all six regional benchmark indices (S&P 500, Russell 2000, FTSE 100, EURO STOXX 50, Nikkei 225, S&P/ASX 200) evaluated from January 2002 to January 2026. All configurations enforce the 0.50% intraday stop-loss and account for institutional round-trip futures transaction frictions.

The decoupled performance metrics provide three crucial insights into the strategy's mechanics:

- **The Destructive Drag of Unfiltered Short Positions:** The unfiltered Short-Only model delivers an inferior CAGR (8.61%) and a Sharpe ratio of 0.810, while suffering an acute maximum drawdown of −31.45%. Taking short contrarian positions against secular bull markets introduces severe frictional drag, leading to prolonged capital decay.
- **Risk Compression via Macro Trend Conditioning:** Conditioning short execution on the SMA_{200} filter repairs the short leg. While CAGR settles at 7.30% due to cash allocations during prolonged expansions, annualized volatility drops to 7.99% and maximum drawdown is compressed from −31.45% to −14.25%. Simultaneously, the trade-level Profit Factor advances

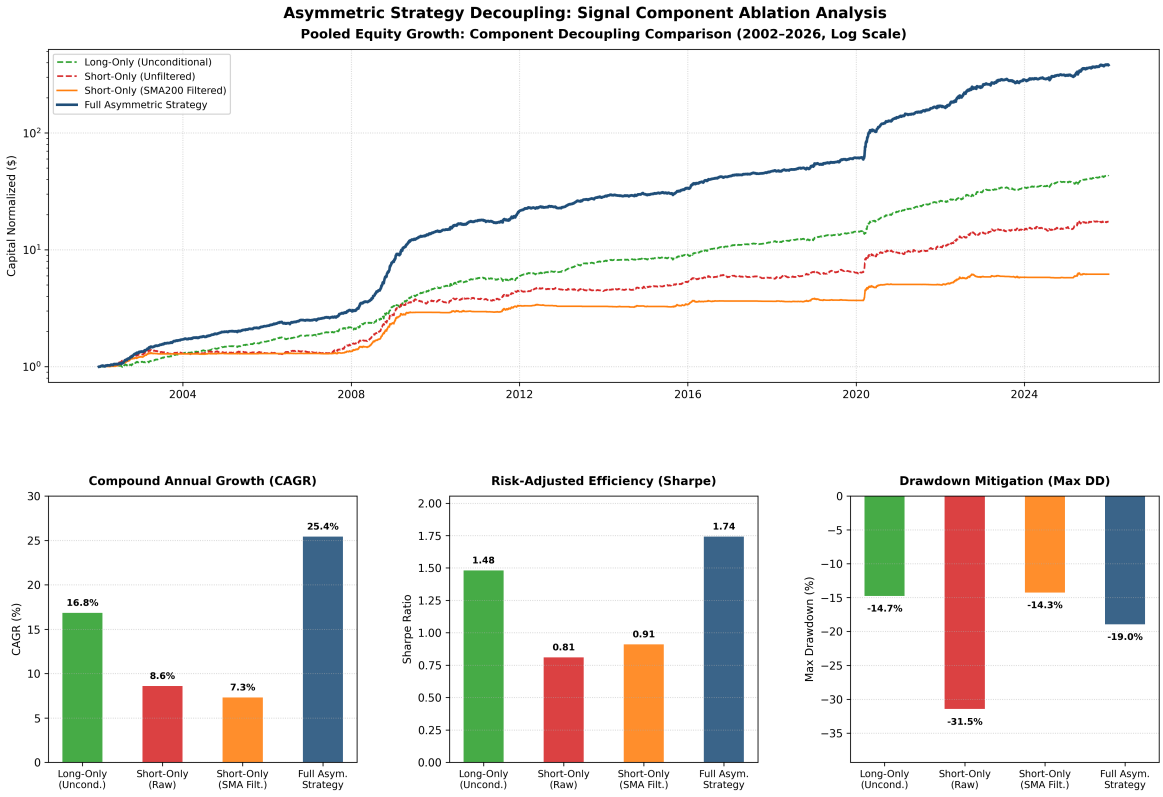


Figure 5. Asymmetric Strategy Decoupling: Signal Component Ablation Analysis (Pooled Global Indices, 2002–2026). The upper panel displays cumulative logarithmic equity curves across decoupled components net of calibrated transaction frictions. The lower panels illustrate comparative performance metrics: (a) Compound Annual Growth Rate (CAGR), (b) Sharpe Ratio, and (c) Maximum Drawdown (%).

from 1.235 to 1.575, demonstrating that short mean-reversion trades are highly effective when restricted to regimes where secular upward drift is suspended.

- **Super-Additive Portfolio Convexity:** Rather than diluting performance, integrating the conditioned short leg alongside unconditional long entries generates powerful diversification synergy. The full asymmetric model achieves a pooled CAGR of 25.43% and elevates the Sharpe ratio to 1.742, significantly outperforming the Long-Only baseline (1.481). The conditioned short trades provide vital crisis alpha during major market panics (e.g., 2008 and 2020), counteracting long-side drawdowns and lifting the Calmar ratio to 1.653.

A.4 Total Transaction Cost Resilience Surface and BETC Validation

To evaluate the operational robustness of the trading signals under institutional execution conditions, we subject both strategy variants to an empirical cost decay analysis. Specifically, we vary the total round-trip transaction cost per trade (c_{trade}) continuously across the domain:

$$c_{\text{trade}} \in [0, 15] \quad \text{basis points per round-trip trade,} \tag{26}$$

where each transaction leg bears exactly half of the round-trip friction ($c_{\text{leg}} = c_{\text{trade}}/2$).

Figure 6 illustrates the comparative compound annual growth rate (CAGR) response surfaces across all six regional benchmark indices and the pooled cross-sectional average for: (a) the Symmetric baseline, and (b) the Asymmetric strategy.

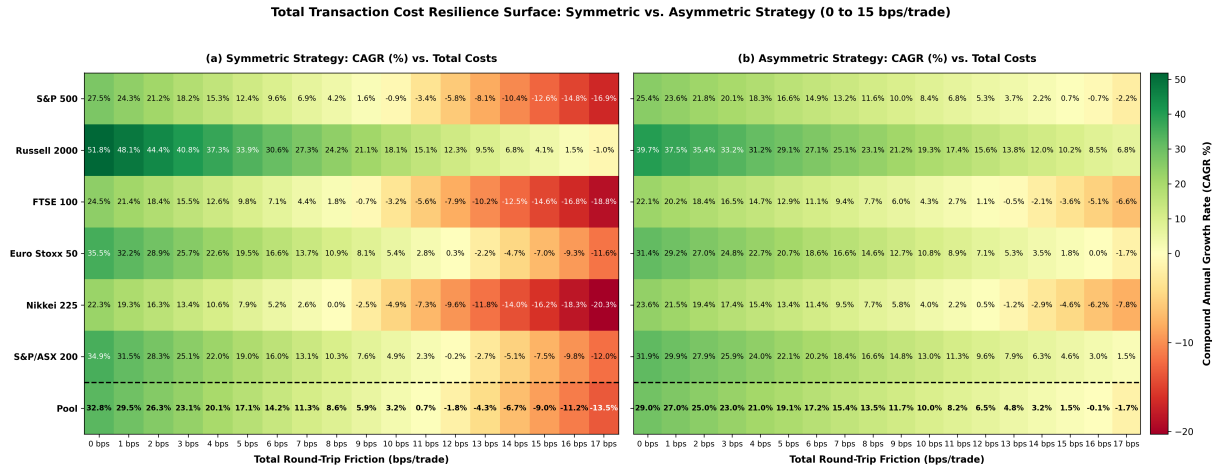


Figure 6. Total Transaction Cost Resilience Surface: Symmetric vs. Asymmetric Strategy (Pooled Global Indices, 2002–2026). Comparative CAGR (%) degradation under escalating round-trip transaction costs. Panel (a) reports the performance surface for the Symmetric baseline; Panel (b) reports the performance surface for the Asymmetric strategy. The dashed line demarcates the cross-sectional pooled average.

The empirical cost surfaces corroborate the analytical Break-Even Transaction Cost (BETC) framework established in Section 4 and reveal the fundamental microstructural advantages of asymmetric execution:

- **Disproportionate Compounding Friction Drag in the Symmetric Baseline:** As evidenced in Figure 6(a), the Symmetric strategy experiences severe geometric decay as transaction costs increase. Gross of frictions (0 bps), the pooled Symmetric baseline generates a CAGR of 32.8%. However, because it executes an unpruned 5,790 trades (≈ 241 trades/year), its performance decays by 41.8 percentage points, plunging into negative territory at 12 bps (−1.8%) and collapsing to −9.0% at 15 bps. For individual indices such as the FTSE 100 (−14.6%) and Nikkei 225 (−16.2%), the rapid onset of capital depletion underscores the operational fragility of unfiltered high-turnover execution.

- **Empirical Confirmation of BETC Break-Even Points:** The empirical heatmap curves reconcile directly with the theoretical metrics reported in Table 6. For the S&P 500 baseline, the analytical BETC relative to Buy & Hold is 7.84 bps/trade. In Figure 6(a), the S&P 500 CAGR degrades from 27.5% at 0 bps to 6.9% at 7 bps and 4.2% at 8 bps, crossing the empirical S&P 500 Buy & Hold rate (≈ 7.5%) precisely at the theoretical BETC threshold. Similarly, the pooled Symmetric BETC threshold of 11.04 bps/trade corresponds exactly to the boundary where excess returns over passive indexing vanish (0.7% at 11 bps versus a long-run passive benchmark).

- **Turnover Pruning and Frictional Immunity in the Asymmetric Architecture:** By conditioning short trades on the macro SMA_{200} filter, the Asymmetric architecture prunes 36.64% of execution volume cross-sectionally (reducing total trades from 5,790 to 3,669, or $\approx$ 153 trades/year). As demonstrated in Figure 6(b), this reduction in portfolio turnover attenuates the compounding friction drag. The pooled Asymmetric strategy preserves positive absolute CAGR throughout the entire evaluation domain, maintaining +1.5% even under extreme frictions of 15 bps/trade (+10.2% on the Russell 2000, +4.6% on the ASX 200, and +0.7% on the S&P 500). The total CAGR compression between 0 and 15 bps is limited to 27.5 percentage points (versus 41.8 percentage points for the baseline), formally proving that structural trade filtering provides substantial immunity against realistic market impact and broker commissions.